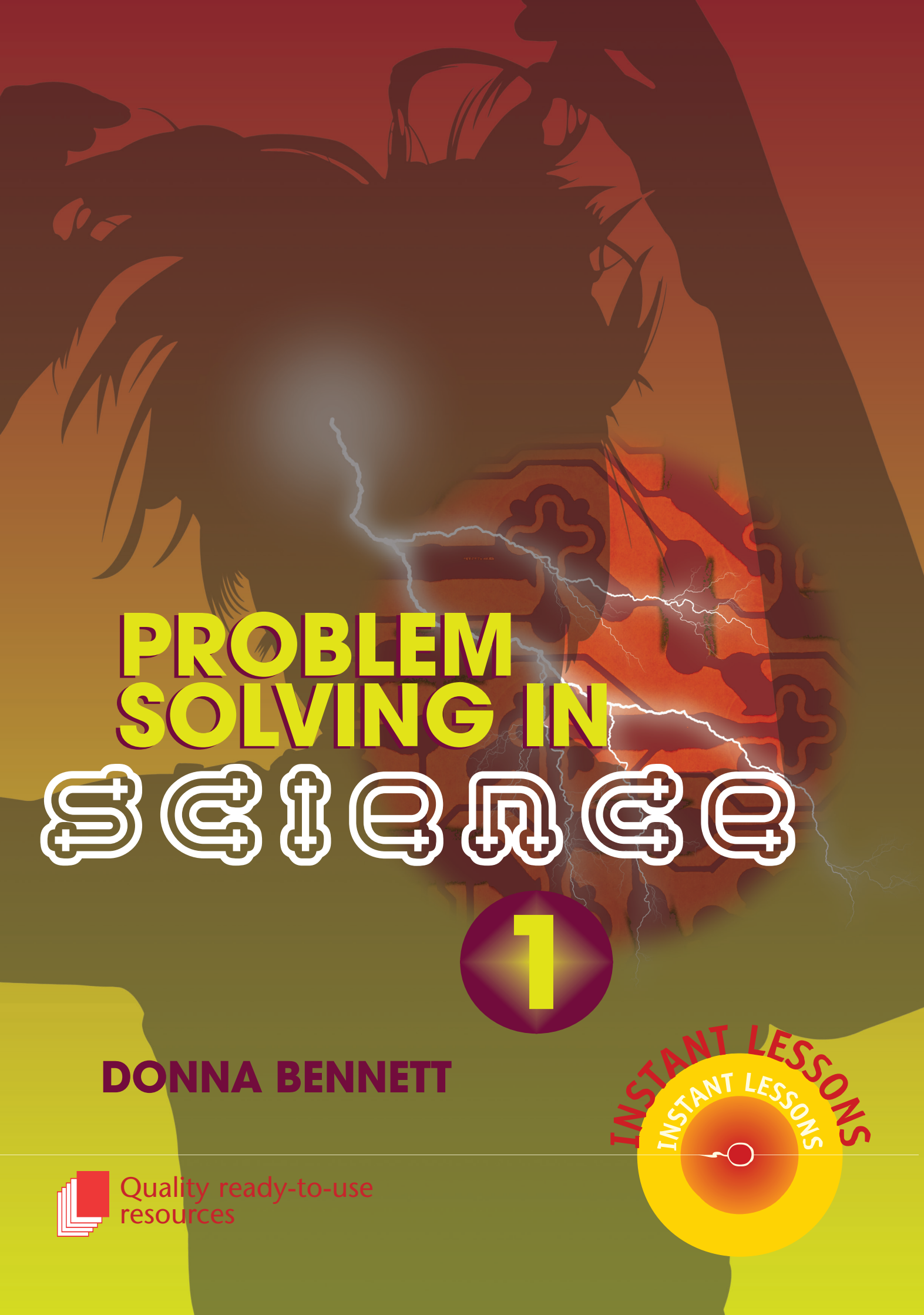




PROBLEM SOLVING IN SCIENCE

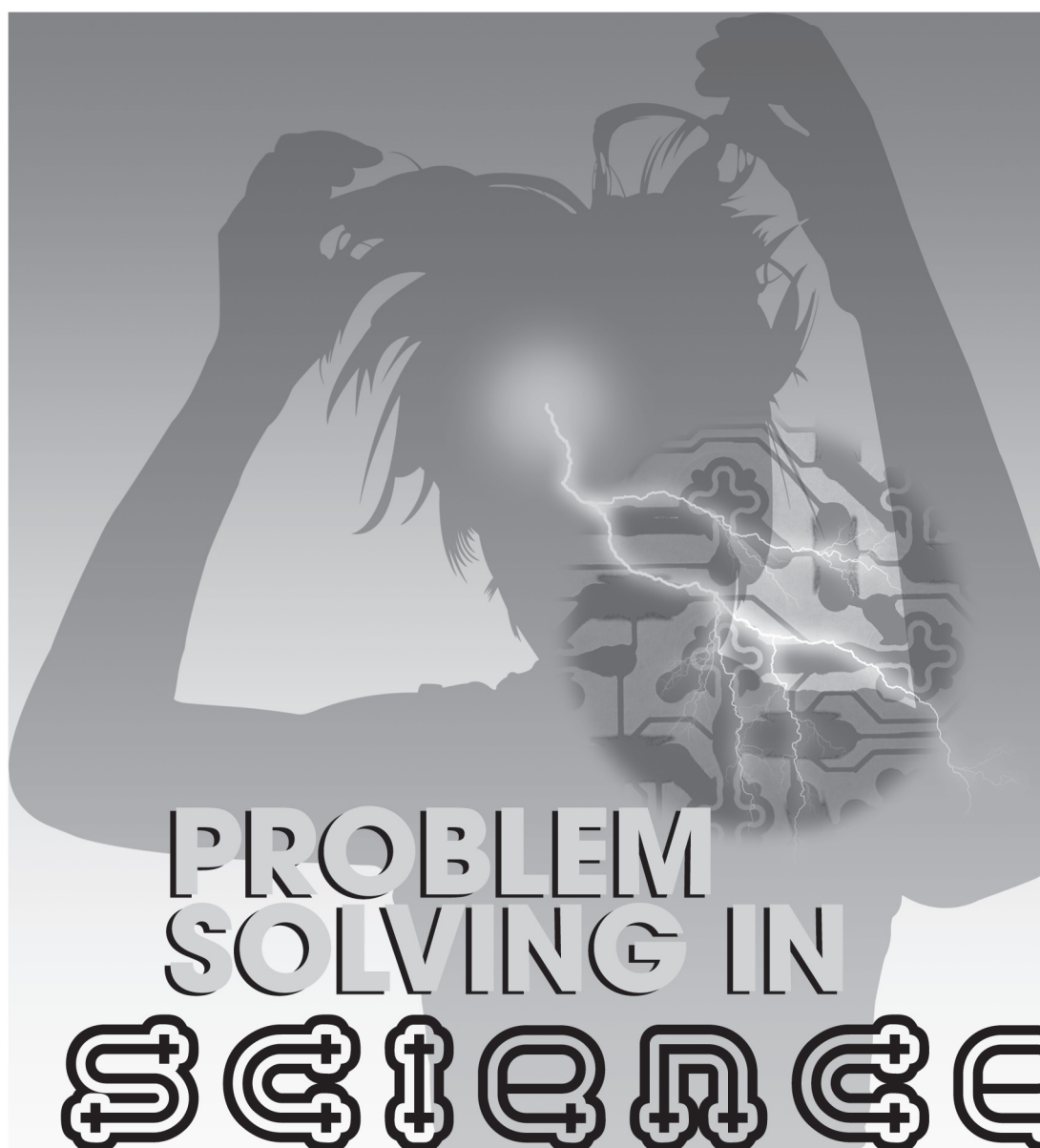
1

DONNA BENNETT



Quality ready-to-use
resources





PROBLEM SOLVING IN SCIENCE

1

DONNA BENNETT





Donna Bennett is a highly qualified and experienced teacher. She is the winner of two national awards, the CRA Mining Company Fellowship and the BHP Science Teacher Award, for innovative and creative delivery of science. She is the Director of Science Education Australia and her science workshop programs are always popular with teachers and students alike. Donna has written several successful science education books, established the Queensland Science Marathon and has taken her 'Science for Bush Kids' program to regional areas. She has travelled internationally where her special form of science education has been in demand.

Donna believes that science is the greatest adventure that has ever been devised. She sees teachers as consultants and cheerleaders who provide probing questions, enthusiasm and confidence and who tease out an understanding of the basic notions of science. Teachers are a source of knowledge and information, which is shared in order to guide student exploration, rather than define it. As an author she has a clear and effective style and believes that interactive science investigations contribute most to learning when they are part of a well-thought out plan that triggers the natural curiosity of students. She provides a freedom that allows students the pleasure of building their own knowledge and understanding. Once they get over the shock of having this type of responsibility, and experience the thrill of succeeding, nothing can stop them.

© Donna Bennett and Blake Education 2007

Copying of this book for educational purposes

The purchasing educational institution may only photocopy pages within this book in accordance with The Australian Copyright Act 1968 (the Act) and provided the educational institution (or body that administers it) has given a remuneration notice to the Copyright Agency Limited (CAL) under the Act. It is a breach of copyright to scan or in any other way make digital copies of the worksheets.

It is mandatory that ALL photocopies are recorded by the institution for CAL survey purposes.

For details of the CAL licence for educational institutions contact:

Copyright Agency Limited
Level 19, 157 Liverpool St
Sydney NSW 2000

The Act does not permit the purchasing educational institution to store any part of this book in a retrieval system, or transmit any part of it in any form by any means without the prior written approval of the publisher. All enquiries should be made to the publisher at the address below.

Commissioning editor: Jennifer Gough

Managing editor: Ruth Schultz

Cover design: Gail McManus Graphics

Editor: Catherine Page

Illustrator: Boris Silvestri

Layout and computer art: Final Page Editing and Design

Printed in Australia by Ligare Book Printer

Blake
EDUCATION
Better ways to learn

Instant Lessons

An imprint of Blake Education Pty Ltd

ABN 074 266 023

108 Main Road

Clayton South VIC 3169

Ph: +61 3 9558 4433

www.blake.com.au

ISBN 978-1-92561-006-2

INTRODUCTION

The formulation of a problem is often much more essential than its solution, which may be merely a matter of mathematical or experimental skill. To raise new questions, new possibilities, to regard old problems from a new angle, requires creative imagination and marks real advances in science. (Albert Einstein)

This book helps students to become critical thinkers and independent, creative problem solvers by stimulating student thinking beyond factual recall and procedural levels. This book engages students in higher order thinking, which involves the application, synthesis and evaluation of knowledge and skills within a scientific context. For example, students will not only be able to recognise Newton’s Second and Third Laws but will understand them and be able to apply them to solve a previously unseen problem.

This book is not a laboratory manual or a series of lesson plans. This book is not a bunch of activities and cookbook directions to prove what students already know. However, there is a five-step order of presentation within each unit, and the teacher may interpret and adapt material to fit individual teaching agendas. Each unit begins with a ‘Curiosity trap’, which is located in the teacher’s notes. It is recommended that this be used to grab students’ attention and arouse their curiosity before they know what the teacher has planned. This is followed by the ‘Give us a clue’ section, a hands-on activity to flesh out the basic scientific notion presented by the ‘Curiosity trap’. Next, ‘The challenge’ provides the vehicle to apply this notion to an unknown situation; it is not to be used as an example of the scientific method. Content (‘So what’s going on?’) plus questions (‘Figure it out!’) and answers are provided for each unit, to consolidate an understanding of the basic notions presented.

Using this material for a class or year level competition – A criteria sheet, which outlines a scoring system, is provided on page 5. There are nine criteria that are generic to all challenges, but the tenth criteria is specific to each challenge. This tenth criteria for each unit is provided on page 6.

CONTENTS

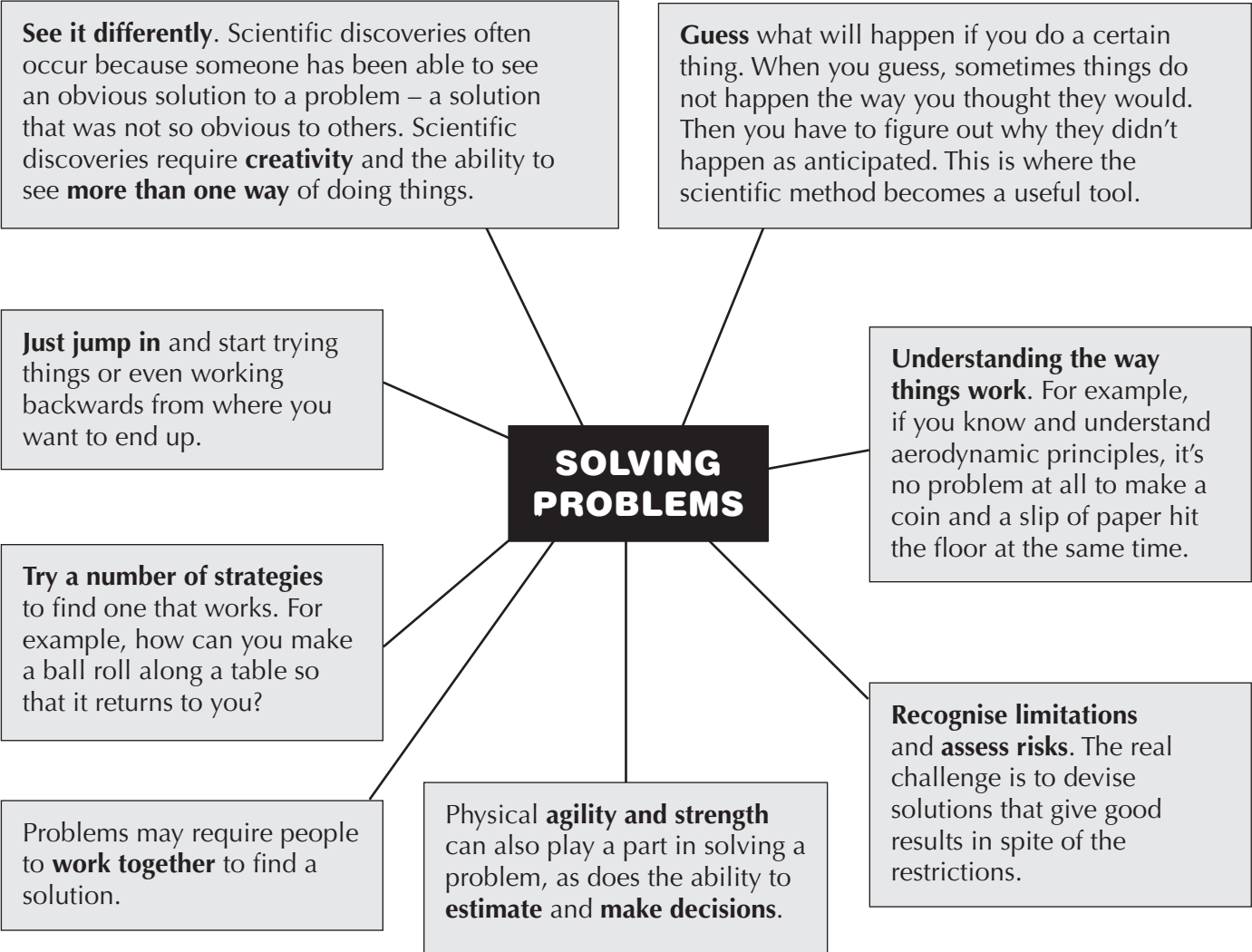
Criteria sheets.....	5
1 Film canister cannons <i>Getting a grip on Newton’s first and second laws of motion through projectiles and catapults.....</i>	7
2 Reaching for the sky <i>Bernouli’s Principle and things that fly.....</i>	12
3 Burglar alert <i>Understanding the electrical nature of matter by unlocking the ‘mystery’ of the electrical circuit</i>	17
4 Keep the heat <i>The kinetic theory of matter in terms of what’s ‘hot’ and what’s ‘cold’</i>	23
5 Robust rafts <i>Balanced forces (floating), unbalanced forces (sinking) and Archimedes’ Principle.....</i>	28
6 Is it science or is it magic? <i>What Newton has to say about the ‘lazy’ nature of matter and the rules of moving anything at all.....</i>	33
7 Under pressure <i>The atmosphere according to Charles and Boyle.....</i>	39
8 Spaghetti science <i>The connections between Newton, structural forces and the manipulation of materials.....</i>	46
9 Daredevil eggs <i>Collisions and the conservation of energy and momentum.....</i>	52
10 Plastics to the test <i>Separating mixtures by searching for the differences.....</i>	58

SCIENCE INVOLVES PROBLEM SOLVING

The scientific method and problem solving can stand alone, but knowledge and skills cannot. Knowing something or knowing about something leads to the raising of a question. The Scientific Method is a great tool to use to find out the answer to the question. Once this new knowledge is accepted, it can then be used to help solve the original problem. We learn nothing unless we proceed from the known to the unknown.

Skills have limited problem solving value in the absence of the ability to use them knowledgeably, just as knowledge has limited problem solving value in the absence of the ability to apply it skilfully. Day-to-day learning should ensure that knowledge and skills are gained simultaneously.

When you want to solve a problem, you try to remember things that you know or things you can do that might help. You look for new information about the question. You may also ask yourself what you know about the situation, what more you need to know, and what assumptions you are making.



The last word

When you try to solve a problem, you try to remember things that you know or can do that might help you. You look for new information about the question. You may also ask yourself what you know about the situation, what more you need to know and what assumptions you are making. Being wise is not only about knowing and understanding lots of things, it's also about applying them appropriately.

CRITERIA SHEET

Teacher's notes

TEAM: _____ **ROOM:** _____ **FINAL SCORE:** ____ /100 **JUDGE:** _____

To score: For each criteria, circle ONE score only, plus the applicable penalties. Then transfer the circled scores to the subtotal. Subtract the total penalties from the initial score and record the actual score in the box. To calculate the final score, add up all the scores in the boxes.

Criteria 1: Team cooperation	Score	Penalty
All team members working cooperatively all the time	10	
All team members working cooperatively most of the time	7	
All team members working cooperatively sometimes	5	
One team member dominating		2
One team member was left out / did not take part		2
Team displaying uncooperative behaviour		2
Subtotal		
TOTAL (score – penalty)		

Criteria 2: Systematic planning	Score	Penalty
Demonstrating strong systematic planning	10	
Demonstrating systematic planning at all times	7	
Little systematic planning	5	
No systematic planning demonstrated		2
Subtotal		
TOTAL (score – penalty)		

Criteria 3: Knowledge	Score	Penalty
Strong evidence of useful knowledge being used	10	
Some evidence of knowledge being used	7	
Little prior knowledge being used	5	
No evidence of prior knowledge being used		2
Subtotal		
TOTAL (score – penalty)		

Criteria 4: Applying strategies	Score	Penalty
Consistently displaying use of strategies to trial and test ideas	10	
Sometimes displaying strategies to trial and test ideas	5	
No clear display of strategies to trial and test ideas (ad hoc approach)		2
Subtotal		
TOTAL (score – penalty)		

Criteria 5: Manipulating and using equipment	Score	Penalty
Excellent manipulation skills	10	
Sound manipulation skills	5	
Inappropriate use of equipment		2
Subtotal		
TOTAL (score – penalty)		

Criteria 6: Safe techniques	Score	Penalty
Demonstrating safe and responsible techniques at all times	10	
Demonstrating safe and responsible techniques sometimes	5	
Demonstrating unsafe or irresponsible techniques		2
Subtotal		
TOTAL (score – penalty)		

Criteria 7: Innovation	Score	Penalty
Demonstrating innovative/creative thinking in solving problems	10	
Demonstrating ordinary/simplistic ways of solving problems	5	
Subtotal		
TOTAL (score – penalty)		

Criteria 8: Attention to task	Score	Penalty
Focused and persistent; constantly on task	10	
Some interruption to concentration	5	
Distracted and inattentive (one or more team members)		2
Subtotal		
TOTAL (score – penalty)		

Criteria 9: Application of basic scientific notions	Score	Penalty
Application of the relevant basic scientific notion	10	
Some evidence of application of the relevant basic scientific notion	5	
No evidence of application of the relevant basic scientific notion		2
Subtotal		
TOTAL (score – penalty)		

CRITERIA 10 SCORING

Teacher's notes

Criteria 10: These criteria are specific for each challenge. Use in combination with the 'Answer' section for each unit.

1. Film canister cannons	Score	Penalty
System 1 (bullet empty)	10	
System 2 (bullet with 30 mL water)		2
System 3 (bullet containing 1 sinker)		2
Subtotal		
TOTAL (score – penalty)		

2. Reaching for the sky

Flight time: The team with the longest flight time will receive 10 points. Slower teams will lose 1 point for every 2 seconds they are behind the longest time. Record the time for each flight attempt within the 3-minute time limit.

	Score	Penalty
<i>Attempt 1 time:</i>	10	
<i>Attempt 2 time:</i>		
<i>Attempt 3 time:</i>		

3. Burglar alert	Score	Penalty
Able to build a circuit that works on the first trial	10	
Able to build a circuit that works on the second trial	6	
Able to build a circuit that works on the third trial	3	
Unable to build a circuit that works		2
Subtotal		
TOTAL (score – penalty)		

4. Keep the heat	Score	Penalty
Temperature change of <5°C	10	
Temperature change of 5–10°C	5	
Temperature change of 10–15°C	2	
Temperature change of >15°C		2
Subtotal		
TOTAL (score – penalty)		

5. Robust rafts	Score	Penalty
Between 5 and 10 washers	2	
Between 11 and 20 washers	4	
Between 21 and 30 washers	6	
Between 31 and 40 washers	8	
Between 41 and 50 washers	10	
More than 50 washers	20	
Less than 5 washers		2
Subtotal		
TOTAL (score – penalty)		

6. Is it science or is it magic?	Score	Penalty
41–50 cubes	10	
31–40 cubes	8	
21–30 cubes	6	
11–20 cubes	4	
1–10 cubes	2	
0 cubes		2
Subtotal		
TOTAL (score – penalty)		

7. Under pressure	Score	Penalty
Egg removed from the bottle without any breakage	10	
Using a valid solution but egg damaged in the process	2	
Unable to remove the egg from the bottle		2
Subtotal		
TOTAL (score – penalty)		

8. Spaghetti science	Score	Penalty
Structure held three books off the table for 30 seconds	10	
Structure held two books off the table for 30 seconds	5	
Structure collapses with the weight of two books		2
Subtotal		
TOTAL (score – penalty)		

9. Daredevil eggs	Score	Penalty
Egg does not break	10	
Egg breaks		5
Subtotal		
TOTAL (score – penalty)		

10. Plastics to the test	Score	Penalty
Yes it will float	5	
No it will not float		2
Correct explanation	5	
Partially correct explanation	2	
Incorrect explanation		2
Subtotal		
TOTAL (score – penalty)		

Safety goggles must be worn by all those observing this challenge. No-one must be near the cannon until it fires. If Alka-Seltzer gets splashed in someone's eye, wash the eye out with plenty of water. To avoid unnecessary mess, set this activity up on newspaper.

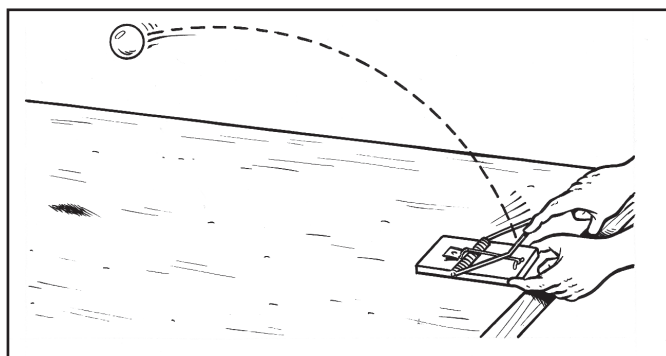
Test-fire the container following the instructions in the challenge. It is important to carry out the instructions *quickly* for the best results. If the container does not pop apart within two minutes, carefully place your hand on top of it, firmly holding the lid and bottom together. Gently lift the lid and release any remaining pressure.

Do not use any fuel other than the Alka-Seltzer. Film processing shops are usually willing to give you film canisters for classroom use – Kodak brand are by far the best. Once they have been used a few times, replace them with new ones. Railway track can be purchased from hobby shops, but any similar tracking may be used. The wooden blocks for the 'Give us a Clue' section should be made according to the diagram (one per group) prior to the day. Polystyrene balls are available at craft stores.

LINKS TO THE EVERYDAY – Rockets, jet engines in aeroplanes, jet boats, fire arms.

★ Curiosity trap

Using a mouse trap, fire a ping-pong ball projectile above the heads of the students. The set mouse trap is a catapult lever arm that is loaded with energy. When the energy is released, the ball is hurled into the air and the mouse trap springs back. These two movements are the results of the equal and opposite forces in Newton's Third Law. Keep your fingers out of the way of the arm; even though injury is not likely, it can cause a fright.



★ Answers

- 1 **a** As the Alka-Seltzer dissolves in water, a gas is produced that provides the force **b** The gas is carbon dioxide
- 2 **a** The cannon and the bullet have the same mass and they move about equal distances. The action and reaction forces are equal, as shown by the equal distances the canisters move **b** They move in opposite directions, because the action and reaction are opposites
- 3 **a** The greater the force on an object, the greater is its change in speed or direction; and the heavier an object, the less its change in speed or direction **b** The mass of the projectile was increased by using bigger balls, and the amount of energy available was changed by using different-sized rubber bands **c** The small marble travelled the furthest
- 4 The same force is acting on both the cannon and the bullet, their masses are the same, so they move the same distance but in opposite directions
- 5 To prevent the cannon jumping back and hitting the person firing it
- 6 Air is sucked into the jet engine and compressed, and fuel is squirted in and burnt continuously. The burnt gases blow out the back, pushing the engine and the attached aeroplane forward – Newton's Third Law
- 7 Newton's Second Law of Motion is involved **a** The more mass an object has, the less it will change speed or direction **b** The greater the force on an object, the greater its change in speed or direction
- 8 **a** Newton's Third Law – forces are always in pairs, and here the two forces are always equal in size but opposite in direction **b** (i) The force of the block of wood being pulled downwards onto the table by the force of gravity and the force of the table pushing upwards on your elbow (ii) Newton's Third Law
- 9 The small child will move most because of the smaller mass
- 10 The 'hidden forces' of air resistance (drag) and gravity will cause the flight path of any projectile to curve around the Earth until it lands. Gravity pulls it down and air resistance slows it down. It will not go on in a straight line forever. This is called curvilinear motion

★ Give us a clue - Catapults

What you need

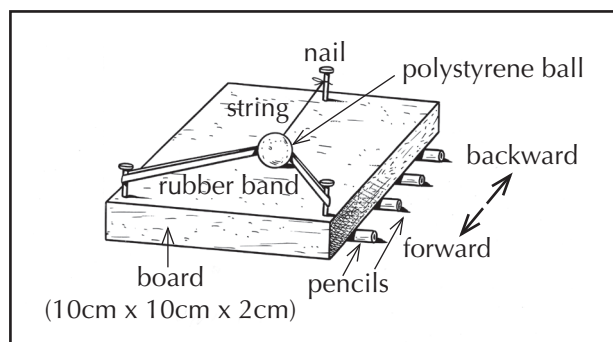
Block of wood with three nails in it, Polystyrene balls (1 and 2 cm diameter), Four round pencils, Rubber bands of different sizes, Cotton, Scissors

SAFETY!

⚠ Do not aim the polystyrene ball at another person.

What to do

- 1 **Set** the catapult down on a smooth surface and tie the cotton onto the nail and the rubber band, as shown in the diagram below.
- 2 **Cut** the cotton and watch the movement of both the ball and the catapult. Observe which direction the catapult travels in, relative to the direction the ball goes.
- 3 **Put** the catapult on the pencils and fire it again.



Now answer these questions:

- How far did the ball travel this time?
- What happened to the block of wood?
- If you change the amount of stretch by using a different-sized elastic band, what happens?
- If you change the mass of the projectile by using a smaller ball, what happens?

What happens next?

After you have completed this catapult experiment and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen and look for clues – stay alert!

★ The Challenge

Within 20 minutes, work out which of the following three systems, designed to fire a bullet from a cannon, would be the safest for the person operating the cannon.

Clues!

A **projectile** is an object fired by force, like a missile, a bullet or a cannon ball.

Mass is a measure of how much matter, or in other words, how many atoms there are in something. Matter is anything that takes up space and can be weighed.

A **catapult** is a 'Y'-shaped device with an elastic strip for propelling objects.

Energy makes things happen, or causes things to happen.

A **system** is an ordered collection of processes with inputs and outputs that stands alone.

Materials

- Two Kodak film canisters, 0.7 m of model railroad track (size 'n'), 1 m tape measure, Alka-Seltzer tablets, 15 mL measuring cylinder, Water, Masking tape, Permanent pen, One medium-sized lead sinker, safety goggles

Instructions

- 1 **Label** one canister 'Bullet' and the other 'Cannon', using masking tape and a pen.

System 1

- 2 **Position** the empty bullet midway along the track.
- 3 **Load** the cannon with 15 mL of water.
- 4 **Add** one-quarter of an Alka-Seltzer tablet to the cannon.
- 5 **Quickly** place the cap firmly on the cannon and hold it in place, then straight away put it on the track so that the tops of the bullet and cannon canisters are touching.
- 6 **Measure** the distances travelled by both the cannon and the bullet and record the results in the table provided on the next page.

Following all the safety instructions:

System 2 – Put 30 mL of water in the bullet, then repeat steps 3 to 6.

System 3 – Put a lead sinker inside the bullet, then repeat steps 3 to 6.

SAFETY!

- ☞ Put on the safety goggles before starting the challenge.
- ☞ Always keep a safe distance from the bullet and cannon when you perform the experiment.
- ☞ If the cannon does not fire within 2 minutes, ask an adult to remove it, then set up another.
- ☞ If Alka-Seltzer gets in your eye, wash the eye with plenty of water.

SAFETY!

- ☞ Step back immediately until the cannon has fired. Remember, if the cannon does not fire within 2 minutes, ask an adult to remove it and then set up another.

★ So what's going on?

The cannon and the bullet are very similar in mass. The empty bullet has slightly less mass than the loaded cannon, but when the top comes off the cannon, the small difference in mass is effectively cancelled out.

*Newton's Third Law:
For every action there is an equal and opposite reaction.*

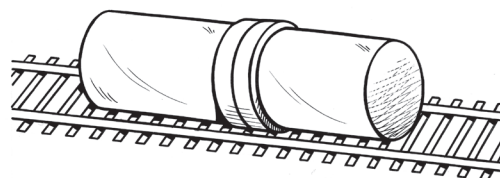
Forces always occur in pairs, and the two forces are always equal in size but opposite in direction.

The force is transmitted to both the cannon and the bullet equally. Since the same force is acting on both the cannon and the bullet and their masses are the same, this illustrates **Newton's Third Law**.

Newton's Second Law: The greater the force on an object, the greater is its change in speed or direction; and the heavier an object, the less its change in speed or direction.

Changing the mass of the bullet by adding water and/or a lead weight demonstrates **Newton's Second Law**. If one of the objects acted upon by a force is much more massive than the other, the distances that the two objects move will be different.

Caps of film containers must touch



A **chemical reaction** is a process in which one or more chemical elements or compounds meet and form new compounds. Evidence of a chemical reaction that can be easily observed is the production of gas. Carbon dioxide gas is produced when the Alka-Seltzer reacts with the water in the cannon.

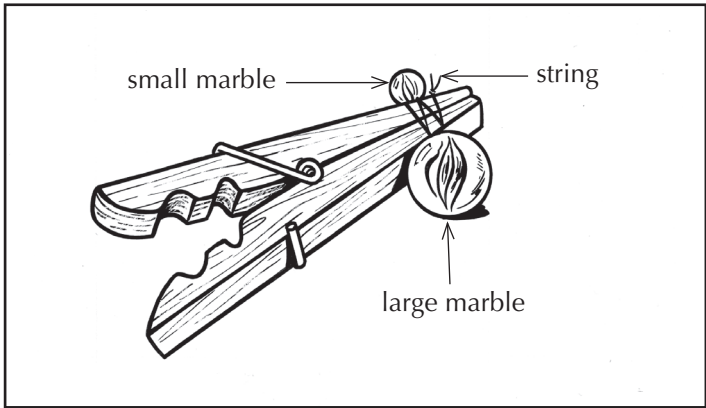
Data table

DISTANCE	EMPTY BULLET	BULLET + WATER	BULLET + LEAD
Distance the cannon travelled	_____ cm	_____ cm	_____ cm
Distance the bullet travelled	_____ cm	_____ cm	_____ cm

The last word: If a real cannon had the same mass as its projectile, the cannon would move backwards towards the people shooting it as fast as the projectile moved forward. There is not much military advantage in shooting a cannonball at someone else if the cannon is going to move backwards and hit you just as hard as the cannonball hits them!

★ Figure it out!

- 1 a What provides the force that moves the cannon and the bullet?
b What gas is produced?
- 2 Consider the work you did in the Challenge, and answer the following questions.
a An empty bullet has about the same mass as the cannon. Compare the travel distance from the starting point of the bullet and the cannon in this case.
b In what directions do they move?
- 3 a State Newton’s Second Law.
b Describe how what you observed and measured in the catapult experiment relates to Newton’s Second Law of Motion.
c A group of students set up the following experiment. They opened a clothes peg and tied it so that it stayed open. They put a small marble on one side and a large marble on the other side, then cut the string. Which marble travelled the furthest?



4 Explain the statement 'For every reaction there is an equal and opposite reaction.' How does it relate to the results you obtained in the challenge?

5 Real cannons are always much more massive than the projectiles they fire. Why must a cannon have more mass than the projectile that it fires?

6 Explain how a jet aeroplane moves forward.

7 **a** Why it is easier to throw a light rock further than a heavy rock?

b Why is it easier to push an empty shopping trolley than a full trolley?

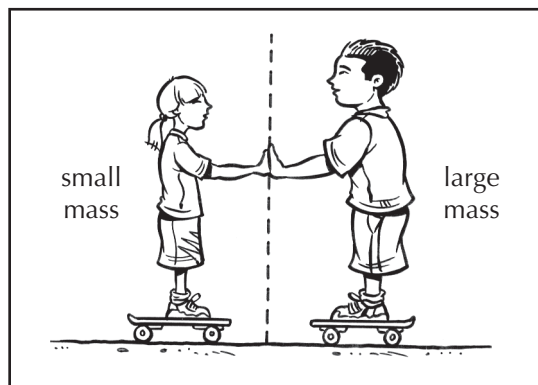
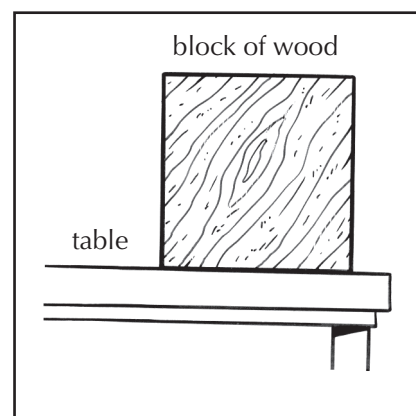
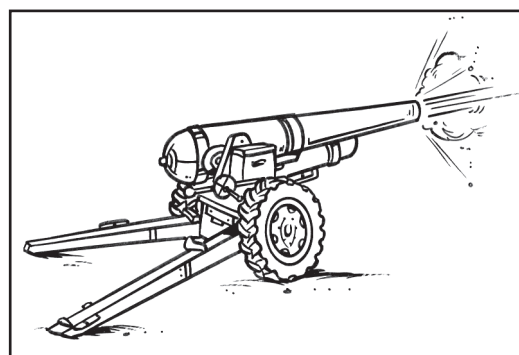
8 **a** Use Newton's Laws to explain what is happening in this situation: two teams are having a tug of war, but the rope is not moving and neither team is moving. What forces are involved?

b Consider the block of wood resting on the table below.

(i) Identify the pair of equal forces.

(ii) Which of Newton's laws is being applied here?

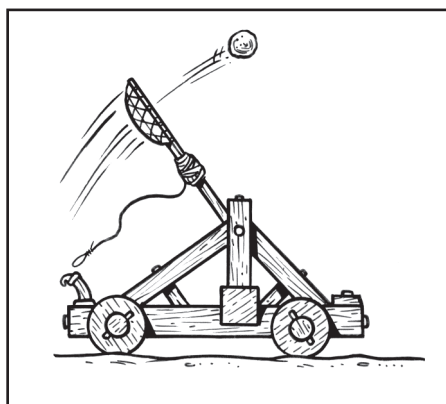
9 Two students, one big and one small, each stand on a skateboard, facing each other with their hands touching. If they push each other apart, predict which student will move the greatest distance.



10 Extend your understanding

Projectiles from cannons and rifles, shot-puts, cricket balls, motorcyclists jumping rows of cars, the Olympic hammer throw and ballet dancers all involve curvilinear motion, because there are forces such as gravity and air resistance operating.

When a catapult or a cannon ball is fired, does it go in a straight line? And does it go on forever?



★ Air support

In Experiment 1, the paper dips between the two books because blowing underneath the paper lowers the air pressure under the paper and the greater air pressure above the paper pushes down on it. In Experiment 2, a spray of water should flow from the first (vertical)

straw. The water rises because the moving air stream from the horizontal straw lowers the air pressure just above the vertical straw and the greater pressure of the surrounding air pushes the water in the glass up the straw.

★ The challenge

A standard school laboratory would provide adequate space for this, but an assembly room or hall would be better. This stunt can be an individual or group challenge. Individual scores can be combined and averaged as a team score. The wingspan must be measured before testing begins. Planes must be hand-launched according to the rules listed under 'The challenge'. Measure the straight-line distance from launch spot to landing spot. Flight time stops when

the device is no longer aloft but is resting wherever it landed; bouncing off walls etc. does not stop the timing.

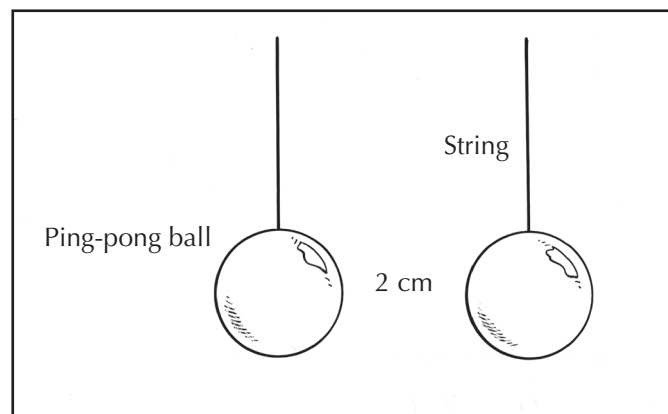
Other tests may include accuracy hitting a target, duration aloft, upward or downward loops, turns around specific objects.

LINKS TO THE EVERYDAY – birds, hang-gliders, seed dispersal, sailing boats, helicopters, golf balls, kites.

★ Curiosity trap

Stick two ping-pong balls onto two pieces of string. Suspend the balls side by side and about 2 cm apart. Take a straw and blow gently into the space between the balls. The straw must be at right angles to the strings and level with the space between the balls. The harder you blow, the closer the two balls swing together, because fast-moving air is flowing between the balls. The pressure therefore drops and the balls are pushed together by the greater air pressure around the outside of the balls.

Before the demonstration, ask the students to predict what will happen when you blow.



★ Answers

- 1 a Pitch b Drag c Thrust d Lift
- 2 a The flame will not go out and will be drawn back slightly toward the centre of the funnel b Pick up the funnel and place it over the ball. Blow through the stem of the funnel whilst picking the funnel up. The ball will come with it. The moving air around the ball exerts less pressure than the still air under the ball c Bernoulli's Principle
- 3 Newton's Third Law and Bernoulli's Principle
- 4 a To reduce air resistance or drag b Shuttle – yes/low; parachute – no/high
- 5 Kites follow the same process as an aeroplane, with a higher pressure on the underside; but there is no thrust, so kites won't fly without a wind
- 6 If aeroplanes move too slowly (not enough thrust), the pressure differences above and below the aerofoil are not great enough to maintain lift and the aeroplane will be pulled to earth by the force of gravity

- 7 a The dimples drag and trap a layer of air over the ball as it spins backwards. The air trapped under the ball is going against the wind, so the bottom air is moving slower than that on top and the ball stays aloft like an aeroplane b Not as far
- 8 The fast-moving air between them lowers the pressure so much that the windows will be pushed outwards by the normal air pressure inside the train
- 9 A good flight needs a good throw and a comfortable, secure grip. Hold it on the bottom, near the front, and throw it straight up as hard as you can. Student explanations will differ and this is OK
- 10 a A helicopter is a flying machine that hovers. Its propeller is a twisted wing, designed so that when rotated by the engine it produces lift but in a forward direction b The wing shape of a boomerang allows both wings to generate lift, and since it is thrown vertically it curves. The top wing is travelling faster through the air than the bottom wing and generates more lift, tending to rotate the boomerang

★ Give us a clue – air support

What you will need

- **Experiment 1:** a sheet of A4 paper and 2 books of the same thickness
- **Experiment 2:** a small glass, water, food colouring, 2 straws, a sheet of newspaper, scissors

What to do

Experiment 1

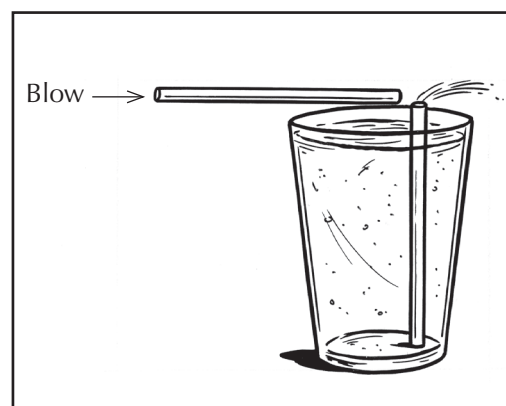
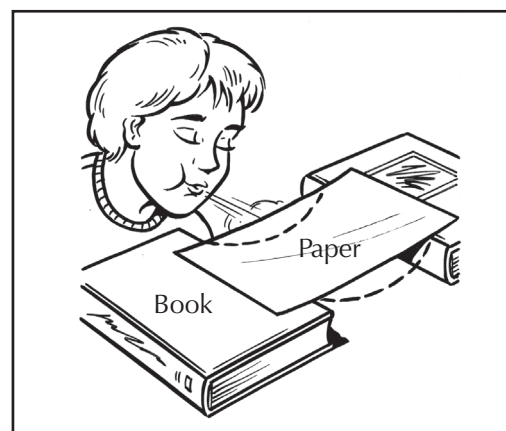
- 1 **Place** the 2 books on the table about 20 cm apart.
- 2 **Put** the paper on top of the books then get down level with the space under the paper and blow gently just above the level of the table, between the books and under the paper.

Experiment 2

- 1 **Fill** a glass with water and add 3 drops of food colouring. Set the glass on a sheet of newspaper.
- 2 **Put** a straw into the water and cut some off its length so that it just extends above the water.
- 3 **Hold** a second straw at right angles to the first straw. While holding the first straw steady, blow through the second straw, across the top of the first straw. Observe carefully what happens.

Now answer these questions:

- What did you notice happened to the paper in Experiment 1?
- Why does the paper behave this way?
- What happened to the water in Experiment 2?
- Why does the water behave this way?



What happens next?

After you have completed Experiments 1 and 2 and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

★ The challenge

Create an aeronautical device that will fly for as long as possible.

Clues!

*Air constantly exerts tremendous **pressure**, but we don't feel it because the pressure is directed in all directions and is balanced by the pressure inside our bodies.*

Aeronautics means the art or science of flying.

An aerofoil is a shaped structure that causes lift when propelled through the air.

Materials

3 pieces of A4 paper, 20 cm of masking tape, 2 paperclips, 2 staples, a child's hula hoop, scissors, tape measure, 30 cm ruler, pencil.

Instructions

You have 15 minutes.

- 1 **Design** an aeronautical device with a wingspan of 7 cm using some, or all, of these materials: three sheets of A4 paper, 20 cm of masking tape, two paperclips, two staples.
- 2 **Give** the device to the judge to have the wingspan checked.
- 3 **Observe** the following rules when testing your device:
 - The wingspan must be 7 cm.
 - All energy given to the device must come from the thrower's arm.
 - Devices must be launched from one hand, with no assistance from other team members.
 - No practice flights are allowed.
 - Both of the thrower's feet must remain inside the hula hoop when launching the device.
 - You will have 3 minutes to fly the device and may fly it as many times as you want to within the 3 minutes. Alterations may be made to the device during the 3 minutes but the timer will not be stopped.

Clue - Why did my plane crash?

Good aeroplanes share three basic types of stability – **pitch**, **directional** and **spiral**.

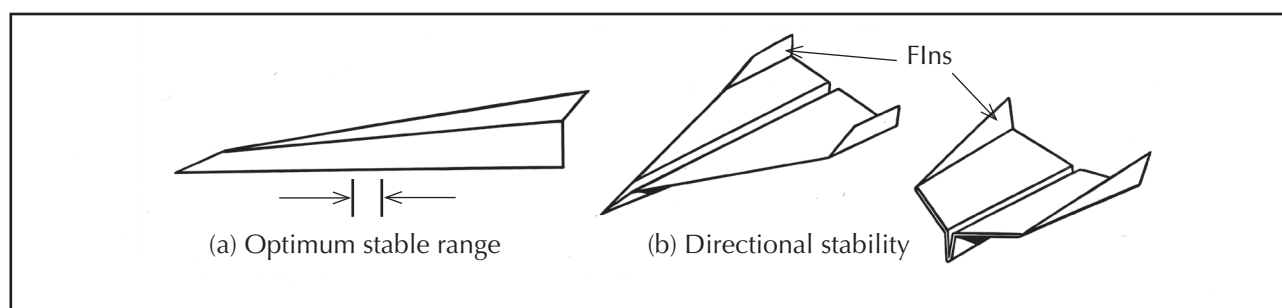
There is a small distance along the length of the plane where it must balance to have good **pitch stability**. This is about half a meter on a 747 and about 2 cm on a paper plane. The best way to tell if your plane has **pitch stability** and therefore glides smoothly is to toss it and see if it dives or swoops up and down. A paperclip on the plane's nose will help with stability.

If your plane has **directional stability**, it will fly in a straight line and will not spin around and

fly backwards. If most of the plane's body is behind the balance point and fins have been provided, the plane should be **directionally stable**.

If a plane is **spirally stable** it will fly in a straight line or a slow constant curve and will not begin to move in tight circles. When looking at the plane from the nose, bend the wings up slightly so that they form a 'Y' shape and make sure they are even.

How do you deal with a real dud? Screw it up and throw it in the bin!

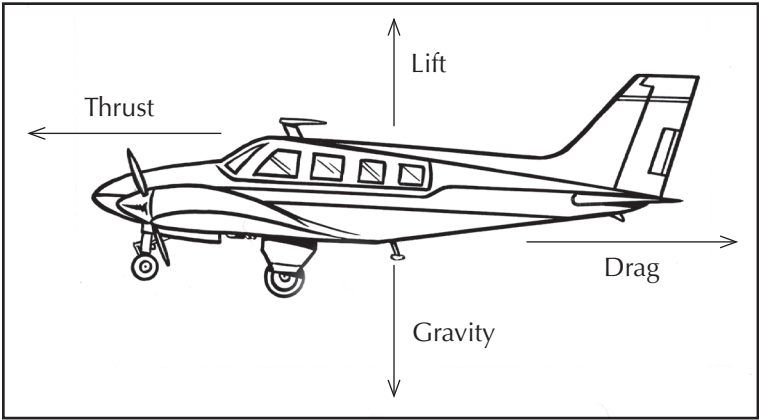


- A judge will time each flight and record the result in the data table below. The longest flight time will be used for scoring.

☆ So what's going on?

Why do aeroplanes fly?

Every aeroplane is in the centre of four forces – lift (upward force), thrust (forward push), gravity (weight) and drag (air resistance = friction caused by air rubbing against the plane as it moves). Lift and thrust must be stronger than the pull of gravity and drag by just the right amount if a plane is to take off. Thrust is usually provided by a propeller or jet engine. The engine is used to overcome the force of gravity and a streamlined design causes air particles to flow smoothly around the plane, reducing drag. When the lift is greater than the weight of the plane, it takes off. An aeroplane will continue to fly if the lift equals the pull of gravity and the thrust equals the drag of air resistance.



Forces on a plane

Newton's Third Law: For every action there is an equal and opposite reaction.

The rounded shape of a wing, called an aerofoil, is designed so that the upper surface is higher than the lower surface. The air therefore flows over the upper surface faster than the lower surface. This results in a lower pressure above the wing, producing an upward force called lift.

As speed increases, so does difference in air pressure and therefore lift. An aeroplane gets lift from pressure differences above and below the wing, together with its speed and the angle of its climb.

The Bernoulli Principle: As the speed of air (or liquid) increases, its pressure drops; and as its speed decreases, its pressure increases. (Daniel Bernoulli was a Swiss scientist who lived in the 18th century.)

The last word: Paper aeroplanes fly using the same general principles as real aeroplanes. However, lift is caused by the 'angle of attack' of the wing. Gravity tugs the plane down and lift pushes it up. Thrust and drag tug the aeroplane forward and aft in opposite directions horizontally. If all these forces are equal then the aeroplane is in straight level flight. Throwing a paper plane provides its initial thrust, then as the pull of gravity and drag balance each other the plane does not slow down but it does descend in a curve.

Data table

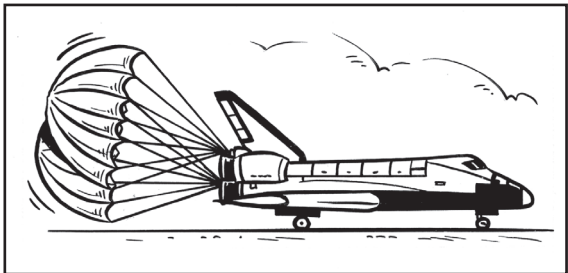
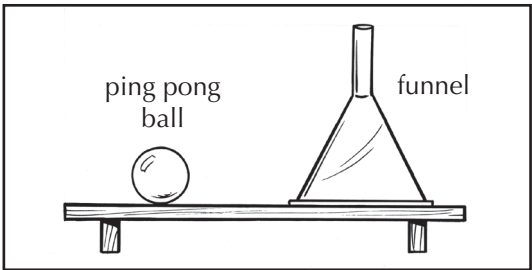
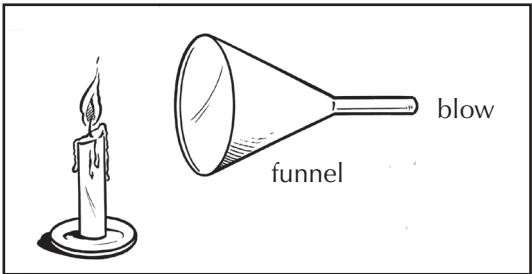
FLIGHT NUMBER	FLIGHT TIME
1	
2	
3	
4	
5	

★ Figure it out!

1 Match the following words to each of the sentences below: drag, lift, pitch, thrust.

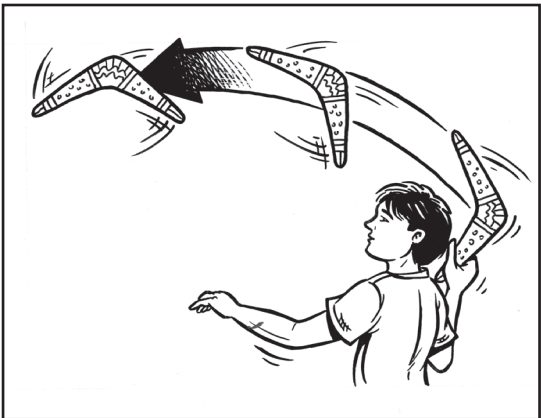
- a _____ is the up and down movement of the nose and tail of an aircraft.
- b _____ is the resistance of the air to moving objects.
- c _____ is the force generated by propellers or a jet flow that propels the plane through the air.
- d _____ is the force generated by an aerofoil when placed in an airstream.

- 2 a Study the diagram provided and predict what will happen to the candle flame when the student blows through the narrow end of the funnel.
- b What will the student have to do to pick up the ping-pong ball with the funnel? They must not touch the ball or suck on the funnel.
- c What principle is being demonstrated by the above two experiments?
- 3 Which two scientific concepts contribute to aeroplane lift?
- 4 a Why are fast cars, jets, birds, rockets etc. especially shaped so that when they are moving fast, air or water flows around them as smoothly as possible?
- b The diagram shows a space shuttle landing. Complete the following table.



	STREAMLINED (YES OR NO)	AIR RESISTANCE OR DRAG (HIGH OR LOW)
The shuttle itself		
Braking parachute		

- 5 Explain how a kite flies.
- 6 What would make an aeroplane stall and fall from the air? Explain in terms of the two pairs of forces acting on an aeroplane in flight.
- 7 a Why does a golf ball have dimples?
- b Would a 'bald' golf ball travel further, or not as far?
- 8 When high speed trains pass each other they must slow down or their windows could be broken. Why? Will the windows be pushed inwards or outwards?
- 9 Write down an explanation of the best throwing technique that will achieve the longest flight time for a paper aeroplane.
- 10 **Extend your understanding**
- a How do helicopters fly?
 - b How do boomerangs fly?



'The shocking truth' unlocks the mystery of an electrical circuit. If students have grasped the concept of a complete circuit and that there are special places

that must be connected by wires (conductors) to both the battery and bulbs or buzzers, they should be able to work out the circuit for the Challenge.

★ The challenge

The biggest problem will probably be keeping all the connections together. The sticky tape will prevent them coming loose. Point out that electrons flow from the negative terminal of the battery to the positive terminal. The black lead on the buzzer must be connected to the negative terminal of the battery. The silver tip on the bottom of the bulb and the thread on the side of the bulb are the special places that must

be connected to the battery. Electronic Alarm Buzzers (70 dB; Cat No: L7009) and light bulbs (Edison Screw Globe; Cat No: 1163Bp1) can be obtained from Dick Smith Electronics.

LINKS TO THE EVERYDAY – zaps from the trampoline, making toast, boiling a kettle, using a computer, turning on a light, switching on the television, power lines from power stations.

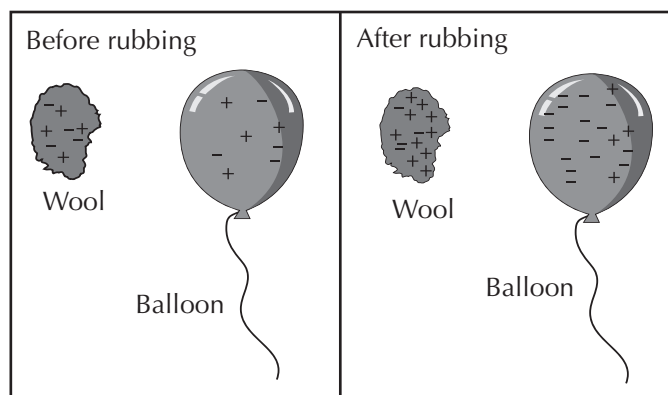
★ Curiosity trap

Take three balloons of different colours, blow them up, tie them off and keep them separate from each other. Tie a piece of thread onto each one. Charge one of the balloons with static electricity by rubbing it hard all over with a woollen cloth. This balloon becomes negatively charged because electrons (which are negatively charged) are rubbed off the cloth onto the balloon.

Static charge

- 1 Using the threads, hold the negatively charged balloon 5 cm away from an uncharged balloon. They will attract each other.
- 2 Hold two negatively charged balloons near each other. They will push each other away, because, as with magnets, like charges repel. Students could be asked what they predict will happen before the activity is performed.

We cannot use static electricity for power as we cannot control the flow of electrons. However, current electricity **will** flow through wires so we can control the flow of electrons.



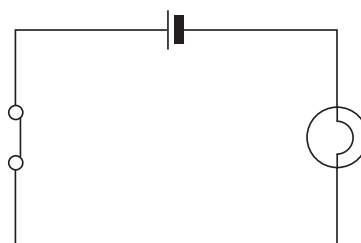
- 3 Connect a 6 volt battery to a buzzer with two electrical wires and demonstrate that electrons generated by the battery will flow through the wires and provide energy to the buzzer as long as the circuit is complete.

Note that it is easier to generate static electricity on cold, dry winter days than on warm, humid days as moisture can pick up charges and quickly discharge the balloon.

★ Answers

- 1 a electron b electric current c electric circuit
- 2 a battery b chemical energy c Copper wire is a conductor and plastic is an insulator, so the electrons cannot escape from the copper wire as the current flows
- 3 a no, yes, no, yes, no, yes b copper, aluminium foil, iron c plastic, paper, wood
- 4 a positive, stays still, static b negative, moves, electric current
- 5 For the bulb to light up, electrons must flow in a complete circuit from the negative terminal to the positive terminal of the battery: A, B, C, H, I
- 6 C

- 7 electrical, sound, light
- 8 Birds and bats can stand on a single power line without any harm but if they somehow touch a second wire at the same time they complete the circuit and get fried as the electrons go through them!
- 9 The diagram should look like the one shown here
- 10 a The circuit would be broken and the light would go out b The current now flows through the lower of the two wires and the light is on



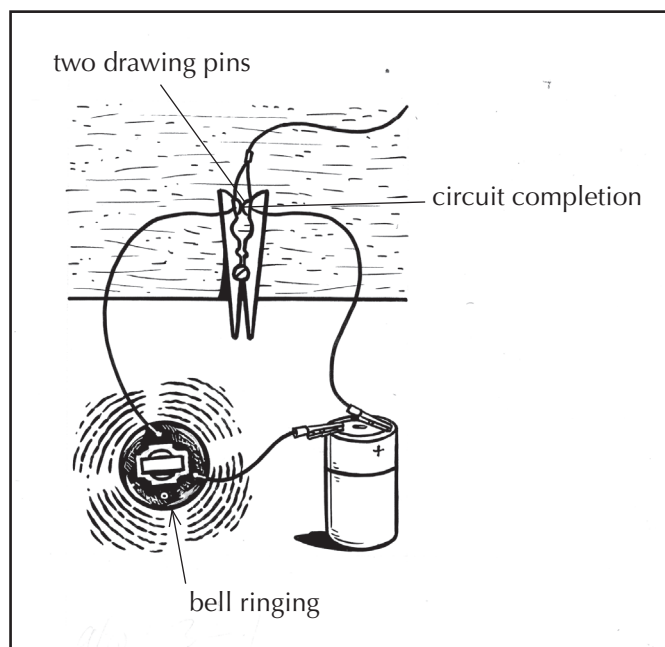
★ Give us a clue – the shocking truth

What you need

6 volt battery, 3 electrical wires with alligator clips, 2 drawing pins, 2 wooden pegs, a little thread

What to do

- 1 **Study** the plan of the electrical circuit drawn here.



- 2 **Build** the electrical circuit using any of the equipment provided. Be careful to follow the plan perfectly. The buzzer will work if you have built a complete circuit. If it doesn't then double-check the plan and make sure that all of the connections are firm and at the right places.
- 3 **Change** this circuit into an alarm.
 - Dismantle the second peg into two halves.
 - Tie a small piece of thread onto the handle of one of the peg halves.
 - Put this half peg between the two drawing pins in the 'jaws' of the first peg. This is the switch.
 - Check that the circuit is still secure. To set the alarm off, pull the thread to remove the half peg. What happens?

Now answer these questions:

- How could you set up this alarm so that you would know when someone goes into your room without asking?
- Using some scrap paper, draw up a plan for building this alarm into the doorway of a room.

What happens next?

After you have completed Experiments 1 and 2 and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

★ The challenge

You have 15 minutes to plan and build an electrical circuit in a doorway so that when a burglar steps on the doormat the people in the house will be alerted.

Clues!

Electricity involves a flow of electrons.

*A **circuit** is a continuous path through which electrons travel.*

*A **switch** is like a bridge for electrons; when you lift the bridge out of the circuit, the electrons can no longer flow.*

Materials

Pen and paper, 2 wooden pegs, scissors, sticky tape, 4 drawing pins, 10 cm of aluminium foil, 6 volt battery, 3 m of electrical wire, 1 bulb, 1 buzzer, 5 paperclips, a thin, light-weight doormat

Instructions

Use any of the equipment provided that helps you to achieve this challenge.

You may **test** the circuit up to three times only before judging, but the best circuit will be the one that works on the first test.

The burglar

The judge is the burglar.



Clues!

Draw up a plan before you begin.

*The **battery** has positive and negative terminals.*

*The **buzzer** has black and red wires.*

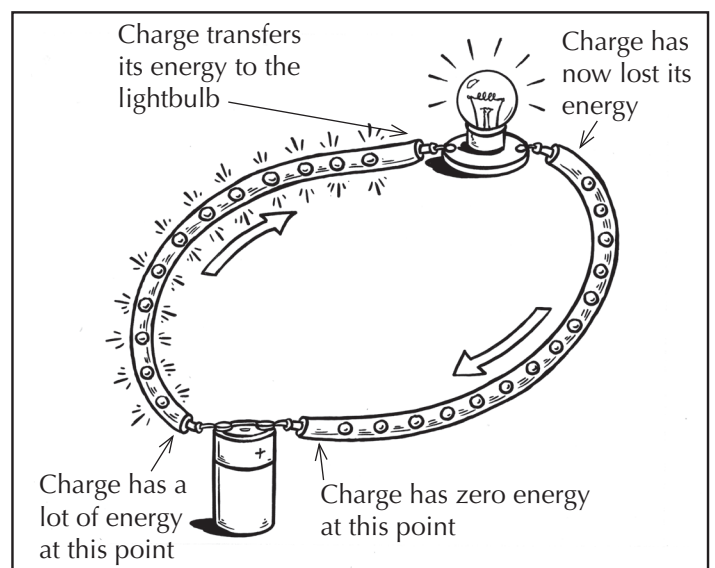
*The **bulb** has a silver tip on the bottom and a metal thread on the side.*

*An **electric current** can flow only if the circuit is made of conductors.*

★ So what's going on?

All matter is electrical

The atoms of all matter are made up of small particles called **electrons**, which are negatively charged (-), and protons, which are positively charged (+). Electricity involves the flow of electrons that carry lots of energy. The electron flow in a circuit using a battery is always in one direction, from the negative terminal to the positive terminal of the battery. Appliances are designed to convert the electron's energy into other forms of energy, such as heat, light and sound.



Electron flow in a circuit

Batteries

Since all matter contains electrons, it is not necessary to 'make' electricity, all you need is something to continually separate the electrons from the protons and push the electrons along a wire. **Batteries** provide the energy to do this. They contain a paste which is either acidic or basic, and two different metal connectors called **electrodes**. A chemical reaction within the battery gives each electron some stored energy, and electricity then flows from the negative electrode around the wires to the positive electrode.

Wires

Copper wires are used in most electric circuits because copper is a very good **conductor**, but electrons flow easily along most metals. The wires are connected to the tops of the electrodes which are called battery **terminals**. Most metals are good conductors of electricity and most non-metals are poor conductors. These poor conductors are called insulators. Electrons do not flow very easily through **insulators**, so they are used to stop electrons going where they are not wanted. Examples of insulators are wood, plastic, air, paper and glass. Insulators can be used as switches to stop and start the flow of electrons.

Circuits

Electrons from the negative terminal of the battery carry energy along the wire until they encounter an appliance such as a buzzer or a bulb. The energy carried by the electrons is then transferred to the buzzer and converted to sound energy, or to a bulb where it becomes light energy. The electrons then flow back to the battery where they receive a new lot of energy and go off around the circuit again. If they cannot get back to the battery because there is a gap in the circuit, everything stops until the circuit is complete again.

Out of control

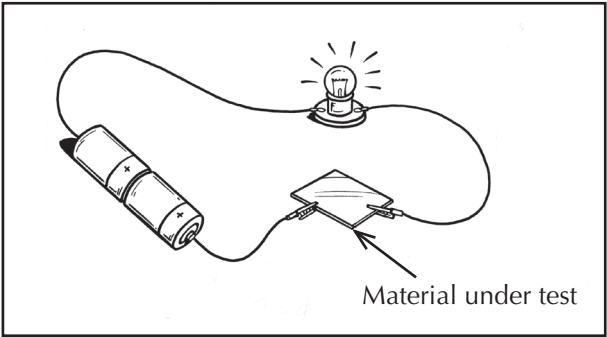
Electrons can be built up on a substance by physically rubbing two unlike substances together so that electrons are rubbed off one substance and transferred to another. Energy is required to separate the positive and negative charges to create this **static charge**. Static electricity will not flow through wires, but when enough charge has built up it will jump across a gap – we feel a zap or see a spark. Lightning is an impressive example. We cannot use static electricity to power appliances because we cannot control the flow of electrons. **Current electricity** will flow through wires, so we can control its flow of electrons.

★ Figure it out!

- 1 Put the correct term into the following sentences: electric circuit, electric current, electron.
 - a An _____ is a negatively charged particle.
 - b The flow of electrons through wires is called an _____.
 - c The pathway that provides for continuous movement of electrons is called an _____.
- 2
 - a What is the name of the energy-providing device that, when placed in a circuit, continually separates negative from positive charges?
 - b What type of energy is used in a battery to separate the charges?
 - c Electrical wires are usually made of copper and covered in plastic. Why?

3 a Imagine doing tests with some materials using the equipment in the diagram. Predict the results, and record them with a 'yes' or 'no' in the table below.

MATERIAL	DOES THE BULB LIGHT UP?
Plastic	
Copper	
Paper	
Aluminium foil	
Wood	
Iron	

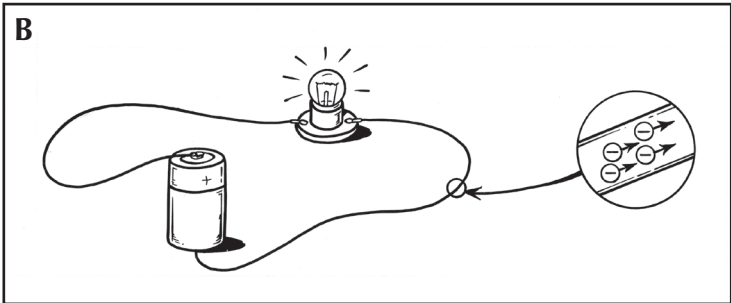
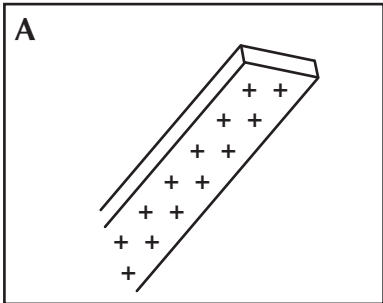


b Which of the materials are metals?

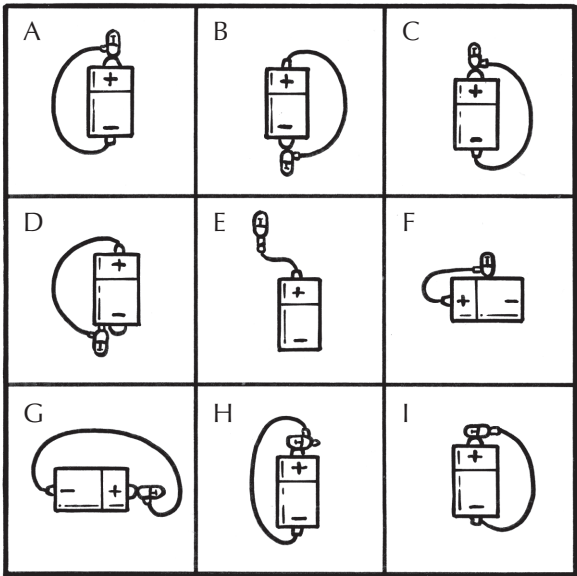
c Which are insulators?

4 Both of the following diagrams show electric charges.
Complete the table below.

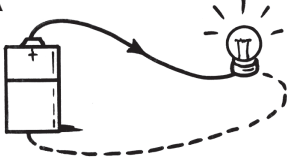
	TYPE OF CHARGE (POSITIVE OR NEGATIVE)	WHAT HAPPENS TO THE CHARGE (STAYS STILL OR MOVES)	STATIC ELECTRICITY OR ELECTRIC CURRENT
A			
B			



5 Study circuits A to I in the diagram below. Tick the ones that will light the bulb.

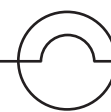
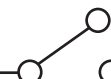


- 6 A battery is connected up to a bulb and the bulb is glowing. Which of the following diagrams best describes the electric current in the wires?

A  No current in dotted wire	B  Current toward bulb in both currents
C  Current the same in both wires	D  Current less in the return wire

- 7 The headlights and the horn of a car work from the battery in the vehicle. Fill in the gaps in the following information.

The battery supplies the _____ energy to the horn, which transfers energy to the surroundings by _____. The headlights are there to transfer energy to the surroundings by _____.

Connecting wire _____	Light bulb/buzzer 
Battery 	Switch 

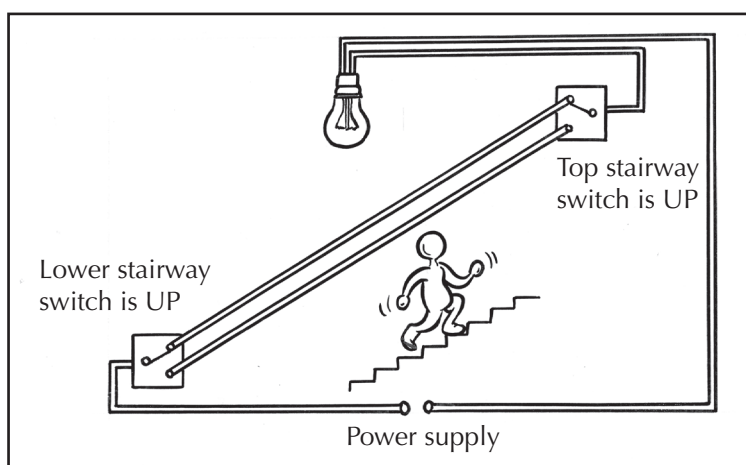
- 8 Why do birds and bats get electrocuted when they stand on power lines?
- 9 Using the symbols shown here, convert the circuit plan that you used to solve the challenge into a circuit diagram. These symbols are recognised by all who work with electricity.

10 Extend your understanding

The diagram shows a two-way circuit that is often used to provide light on stairways. You can switch the light on or off from the top of the stairs or from the bottom. In this diagram both switches are up, the circuit is complete and the light is on.

a If one of the switches was put into the down position, what would happen and why?

b What would happen if both switches were put in the down position?



★ Temperature's rising

The yoghurt containers are the soft plastic type with the reusable lid. Prepare the lids by punching a hole in each lid. Make the hole big enough for

the thermometer to fit through. Alternatively, in a laboratory situation, three pyrex flasks with rubber stoppers could be used.

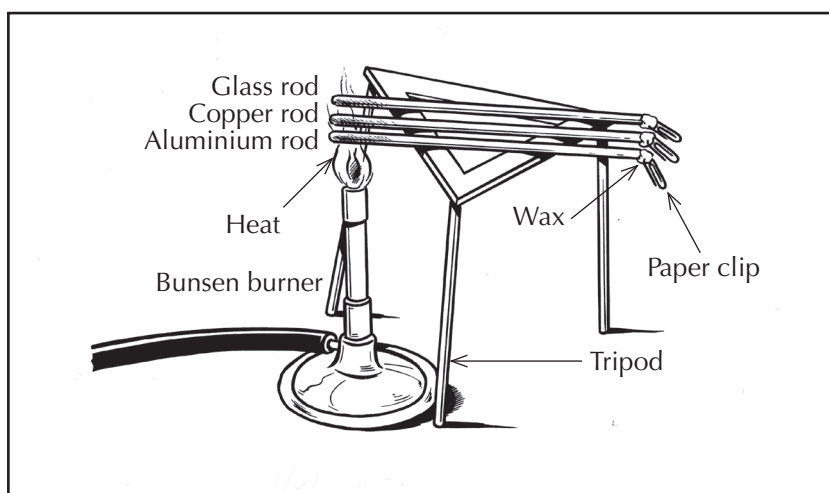
★ The challenge

The judge must measure the temperature at time zero and then at 20 minutes. The smallest change in temperature wins.

LINKS WITH THE EVERYDAY – esky, thermos flask, wetsuit, refrigerator, clothing.

★ Curiosity trap

- Using vaseline, stick a paperclip to the end of a glass rod, a copper rod and a steel rod. Heat the other end of the rods in the flame of a Bunsen burner. Have students predict which paperclip will fall off first and why. The paperclip on the copper rod will fall off first as heat is conducted better by copper than the other two.
- Hold a match head about 2 cm away from a very hot Bunsen flame. Ask students to predict whether the match will catch fire and why/why not. The match will not get hot enough to burst into flame as air is a good insulator.



★ Answers

- Metal is a good conductor. Copper is a very good conductor and efficiently transfers heat to the food; wood is a good insulator and stays cool
- The molecules in the spoon vibrate faster as they absorb the heat. The vibrating molecules keep colliding with other molecules until the energy is transferred from molecule to molecule all the way along the spoon
- a** True (traps air close to the skin) **b** False (a layer of fat is a good insulator) **c** False (it does both) **d** True (traps air close to the bird's skin)
- The person is kept warm by the insulating effects of air bubbles in the rubber foam, by the water trapped between the skin and the suit and by the plastic on the outside of the suit
- Conduction and convection
- All cases maximise the use of air as an insulator by trapping it and preventing body heat from escaping
- Hot, cold, three, radiation, conduction, convection
- a** Glass is a poor insulator and aluminium is a metal and therefore a conductor. **b** Acting as an insulator, the glass confined the heat to a small area, allowing the heat to increase enormously. The aluminium quickly conducted the heat to all areas of the metal, so the heat did not build up enough to burn the wax
- a** Convection **b** Conduction and convection **c** Radiation **d** Conduction
- a** It melts **b** It would melt faster **c** Not enough energy was transmitted through the outside layer of meringue to melt the ice-cream at the centre. The sponge cake and the meringue contain lots of air bubbles, so they are good insulators

★

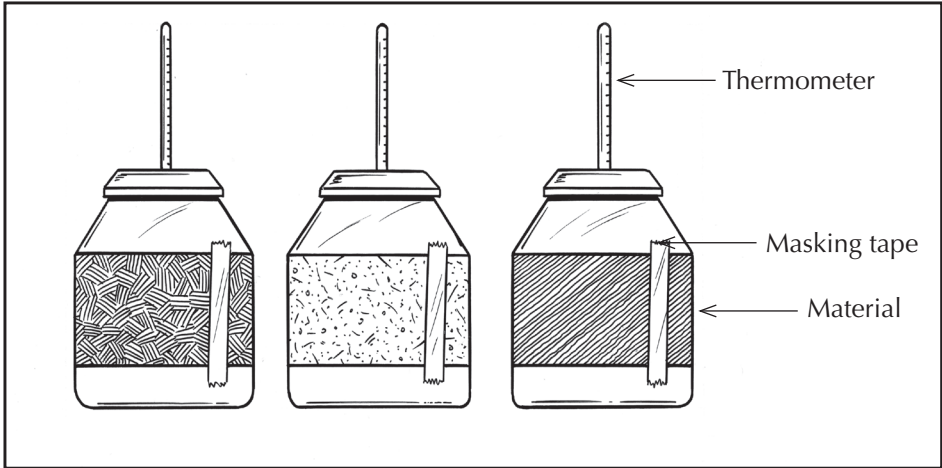
Give us a clue – temperature’s rising

What you need

Three plastic yoghurt containers with lids, magnifying glasses, thermometers, hot water, masking tape, scissors, measuring jug, materials such as wool, cotton, aluminium foil, bubble wrap, newspaper.

What to do

- 1
- Examine the materials with a magnifying glass.
- 2
- Fix the materials to the containers with the tape. Make sure you use the same amount of each material to be sure of carrying out a ‘fair test’.
- 3
- Pour 400 mL of hot tap water into each container and put on the lids.



Yoghurt containers

- 4
- Read the temperature in each container every 10 minutes for 30 minutes.
- 5
- Record the data in the table below.

MATERIAL	TIME ZERO	AFTER 10 MINUTES	AFTER 20 MINUTES	AFTER 30 MINUTES

Now answer these questions:

- Which container lost heat the fastest?
- Which container lost the most heat?
- Which material was the best insulator?
- What was the structure of this material compared to the others you tried?

What happens next?

After you have completed the experiment and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

Heat is one form that energy can take. Hot objects have more heat than cold objects.

*Heat flows easily through a good **conductor**.*

*An **insulator** is a poor conductor of heat.*

★ The challenge

Construct an insulated device which will fit completely inside a 2 litre ice cream container. The device or structure will be designed to retain the heat of 100 mL of boiling water in a standard pyrex 250 mL beaker for 20 minutes.

Materials

2 L ice cream container, 250 mL pyrex beaker, 2 small sheets of newspaper, a 250 mL container of polystyrene pieces, 1 m of masking tape, a roll of toilet paper, 5 small plastic bags

Instructions

- 1 You have **15 minutes** to construct the device, using any of the materials provided.
- 2 After 15 minutes the judge will **impound** the device ready for testing.
- 3 The beaker must be **easily removed** from the device so that the judge may pour boiling water into it and then replace it into the device and measure the temperature change after 20 minutes.

Clues!

*All **gases** are good insulators if they don't move.*

*Gases, liquids and solids are made of atoms that have different amounts of **energy**.*

★ So what's going on?

What's 'hot' – and what's 'cold'?

A substance is hot if its particles have a lot of energy. The particles of a cold object do not have as much energy.

What's heat – and what's temperature?

As a substance takes in heat energy it gets hotter and we say that its temperature goes up. Temperature is a measure of how hot or cold a substance is and **not** how much energy it has. Thermometers are used to measure temperature.

Kinetic theory of matter

All matter is made up of particles called molecules or atoms (in the case of elements) that are way too small to see. All these particles are continually vibrating (but in a fixed place) because they possess energy. The more energy they have, the more they vibrate. As an object is heated, the energy in the vibrating atoms is passed on to nearby atoms, causing them to vibrate with more energy. In some materials, heat (or energy) travels quickly from atom to atom, and in others it hardly flows at all.

Heat transfer

Heat will always flow from hot objects to cold ones. Heat has only three ways to do this travelling.

- Heat is transferred from particle to particle in solids by a process called **conduction**. For example, if you are holding a mug of hot chocolate in your hands, the energy is passed from the hot chocolate to the mug and then to your hands. Water and glass are poor conductors and metals are good conductors.
- **Convection** is where the particles in hot liquids and gases, as they absorb energy, take up more space and rise, such as air in a room with a heater.
- **Radiation** does not need particles to travel; it will travel through a vacuum or through space at the speed of light. Heat flows out from a very hot object, for example from the sun, a radiator or a fireplace. Dull black surfaces absorb the radiation and shiny ones reflect it.

Insulators

Poor conductors are called insulators. Insulators are used to keep hot things hot and cold things cold. We insulate a substance so that very little energy can travel in (if we want the object to remain cold) and very little energy can travel out (if we want the object to remain hot). Some good insulating materials are styrofoam, plastics, wood, newspaper, wool and feathers. Almost all insulators have one feature in common – **air** that is not moving is a very poor conductor and is incorporated into almost all insulators to make them more effective.

★ Figure it out!

- 1 Why are all saucepans made of metal? Why do some saucepans also have copper bases and wooden handles?
- 2 Explain, in terms of heat transfer at molecular level, just why a metal spoon becomes very hot when used to stir a cup of tea or hot chocolate.
- 3 Are the following statements true or false? In each case, explain why.
 - a A fur coat is warmer if worn inside out.
 - b In winter, thin people keep warmer than fat people.
 - c A thermos flask keeps things hot but not cold.
 - d A bird fluffs up its feathers in winter to keep warm.
- 4 Surfers and divers wear wetsuits in cold water. A wetsuit is made of a special foam rubber layer with a plastic waterproof layer on the outside. Water is trapped between the person's skin and the wetsuit. Explain how the wetsuit keeps the person warmer in cold water?
- 5 What form or forms of heat transfer does a vacuum prevent?
- 6 The human body relies on the heat produced by burning food to keep its internal organs at the optimum temperature. What is the one factor common to all of the following?
 - We get goose bumps on our skin when we get cold.
 - We wear several layers of clothing in cold climates.
 - We use a sleeping bag to keep warm.
 - We wear a lined, waterproof suit in the snow.
 - Igloos are made of snow, not ice.

- 7 Read the following and fill in the missing words.

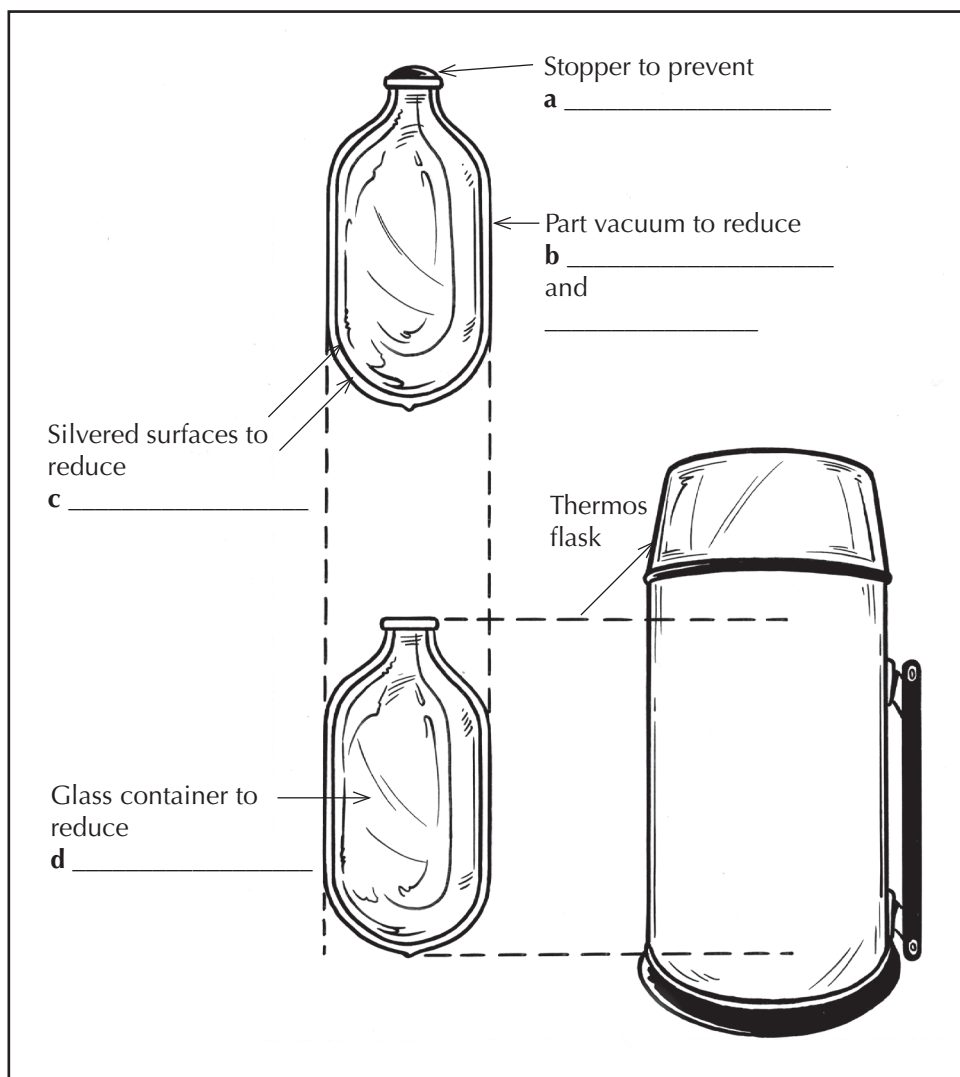
Heat will always flow from _____ objects to _____ ones. Heat has only _____ ways to travel. _____ is a form of heat transfer that does not involve moving molecules. Heat travels in solids by _____ and through liquids and gases by _____.

- 8 Some students are investigating conductors and insulators. They covered a small rectangular strip of aluminium and one of glass with wax and let them cool. They then placed the corners of both rectangles in a flame. They observed that the candle heat caused a small corner of the wax on the glass to actually burn away. However, the wax on the aluminium completely melted but did not burn.

a Which substance was the insulator and which the conductor?

b Why did the wax burn on the glass but not on the aluminium?

- 9 A thermos flask is a glass bottle within another glass bottle. Between the two bottles is a vacuum. The bottles are both lined with a silver coating. The stopper and much of the remaining materials are made of plastic. Using what you know about the transfer of heat, explain how a thermos works by filling in the gaps in the labels.



10 Extend your understanding

- a What happens if you leave ice-cream at room temperature?
- b What would happen to ice-cream if you put it in the oven?
- c Study the plan for making baked ice-cream. Why doesn't the ice-cream melt?

Making baked ice-cream

Ice-cream is put onto a sponge cake base and covered completely with a thick layer of beaten egg white and sugar, which when cooked is called meringue.

The dessert is then baked in a very hot oven for 5 minutes.

The meringue goes brown and the sponge cake gets warm, but the ice-cream remains frozen.

★ Titanic troubles

A cup shape with good sides will be the most successful as it will hold the greatest volume and still float. The volume of the vessel can be calculated by measuring the volume of water displaced when the vessel is floated in water. The ball of plasticine is made less dense by reshaping it so that it contains some air.

This increases the volume of water it displaces and so increases the upthrust or buoyancy force of the water as well as the floatation capacity of the vessel. Salt water is denser than fresh water, so the vessel should hold slightly more weight once the salt is added.

★ The challenge

Washers about 1.5 cm in diameter can be bought cheaply and in large numbers from hardware shops. The judge must check that the three pieces of aluminium foil are 15 cm by 15 cm before construction begins. Instruct the students that they may fold the foil anyway they like, but they must not use any other materials.

After students have constructed their raft, have them predict how many washers it will support without sinking. Record the predictions for each group. The design that holds the most washers is a flat-bottomed

barge type, because the weight of the washers can be evenly distributed over the surface of the water. This type should hold roughly 100 washers. Teams must not test their design before the competition begins because the idea is to think through each design using their knowledge of buoyancy and to make modifications based on any foreseen problems.

LINKS TO THE EVERYDAY – ships, whales, oil slicks, life jackets, travelling coconuts and mangrove seeds, fishing sinkers

★ Curiosity trap

Place a can of regular soft drink and a can of diet soft drink in a deep see-through container of water. The regular can sinks and the diet can floats. The artificial sweetener in the diet soft drink is less dense than the sugar used to sweeten the regular soft drink. The overall density of the regular soft drink is just greater

than water, so it sinks, whereas the overall density of the diet soft drink is just less than water, so it floats.

Introduce some terms involved with carrying cargo across the seas – density, buoyancy, upthrust, weight, balanced forces. What does ‘floating’ mean? What does ‘sinking’ mean?

★ Answers

- 1 **a** displaces, Archimedes, weight, upward, upthrust
b Weight in air minus the weight in water = an upthrust of 2 newtons
- 2 If something stays still, the forces that are acting must be balanced. Water is pushing up, balancing the weight of the whale
- 3 **a** A ping-pong ball floats because it displaces a volume of water that has mass exactly equal to the mass of the ball **b** If you push the ball under the water, the upward force of buoyancy is greatly increased as more water is displaced than if it was just floating on the surface. When you release the ping-pong ball the force of buoyancy causes the ball to pop back up **c** To sink a ping-pong ball in a glass of water you cannot reduce its volume but you can change its mass by sticking more mass onto it
- 4 **a** The hull of a ship is filled with air, and air's density is 816 times less than that of water. If the overall size and weight of the ship are considered, then its density is actually less than that of water and the ship will float **b** 1: Antarctic seas, 2: European river, 3: tropical seas, 4: tropical river **c** The sea in the North Atlantic in a European winter will be much denser than in Singapore, so the ship will float higher
- 5 The molecules of water take up more space as they freeze because they form into a crystal pattern. Ice is therefore less dense than water, so it floats
- 6 **a** The syrup is at the bottom and the surgical spirit at the top **b** It would float between the glycerine and syrup

- layers **c** Oil floats on top of water, so birds get covered in the oil and cannot fly; marine mammals come up for air and get a mouth full of oil
- 7 When a submarine is floating its overall density is less than that of water, just as for a normal ship. When it dives, water is flooded into holding tanks and the density of the submarine becomes greater than water, so it sinks to the required travelling depth
- 8 He got a bar of gold and a bar of silver with exactly the same mass as the crown. He found that the crown displaced less water than the silver bar but more than the gold bar. The crown was not pure gold
- 9 **a** The water level in the dam will fall, because the gravel displaces less water submerged than when it was floating. When floating it displaces only its volume, which is less as gravel is denser than water **b** Lower on the boat
- 10 **a** Measure the volume of water that it displaces. Each mL of water is equal to 1 g. This figure is the mass. Calculate the volume of the block by multiplying length by width by width. Divide the mass by the volume of the block to get the density **b** Weigh the object in air as well as when it is suspended in water and note the difference as this is the buoyant force of the water. Measure the volume of water displaced – this is also the buoyant force. Since the density of water is 1 g/mL, the mass of water displaced also equals the volume of the object in mL. Density = the mass of the object divided by the difference in the two weights (the buoyant force)

★ Give us a clue – titanic troubles

What you need

30 g ball of plasticine clay, 4 L ice-cream container filled with fresh water, uniform weights (such as marbles, beans or coins), paper towel, salt, spring balance

What to do

- 1 **Toss** the ball of plasticine into the container of water. What happens? It sinks, of course.
- 2 **Experiment** with design ideas to make the plasticine ball float.
- 3 **Add** weights (mass) one at a time until the vessel sinks.
- 4 **Modify** the design until it will hold the most cargo possible. Dry and weigh the cargo between tests.
- 5 **Dissolve** salt into the water at the rate of 200 g per litre. What weight (or mass) of cargo will the boat hold now without sinking?

Now answer these questions:

- Does the shape of an object make any difference as to whether it floats or not?
- What is the maximum number of weights that your most successful vessel can carry before it sinks in fresh water?
- Which weighed more: the vessel or the cargo? Does this explain why some vessels sank?
- Which had the most buoyancy: the fresh water or the salt water? Why?
- Which had the greatest density: the ball of plasticine or the plasticine vessel?
- Which had the largest buoyant force: the ball of plasticine or the plasticine vessel?

What happens next?

After you have completed the experiment and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

Buoyancy is the force of displaced water pressing on all sides of a floating object.

Mass is a measure of the how much matter – or in other words, how many atoms – in an object or substance. Matter is anything that takes up space and can be weighed. Mass is measured in kilograms.

Weight depends on the mass of the object and how much gravity is pulling on it.

Density is the mass or number of atoms or molecules in a certain volume. Density can be calculated by dividing the mass of the substance by its volume. The density of water is 1 g/mL.

★ The challenge

Your team has 15 minutes to design and construct the raft that will support the most weight of cargo without sinking in a bucket of water.

Materials

Aluminium foil, scissors, ruler, pencil, bucket of fresh water, paper towels, 100 dry metal washers (1.5–2 cm diameter)

★ Instructions

- 1 **Your team** may build up to 3 rafts. Each raft will be made from one piece of aluminium foil, 15 cm by 15 cm, and the foil must not be cut into smaller pieces.
- 2 **Use** your 3 pieces of foil to experiment with design ideas. Your raft must be watertight and stable in the water.
- 3 **No testing** is allowed before the competition begins.
- 4 **Your team** must decide which one of the raft designs will be used for testing by the judge. Only 1 raft from each team will be tested for competition purposes.
- 5 **Cargo** must not be added to the raft until the judge is ready to supervise. **Your team** must decide which person will load the cargo onto the raft.

To test

The judge will supervise the testing of each team's designated raft one at a time. The test is over when the raft starts to sink – in other words, when water begins to enter the raft.

The raft supporting the most weight of cargo is the winner. Teams are ranked in relation to the winning raft.

Clues!

*The captain of a ship must make decisions about how much cargo to load, as well as how to place the cargo so that his vessel will not sink during the journey. **Archimedes' Principle** tells*

*us that the weight which may be floated is equal to the weight of water displaced by the vessel. This **buoyancy force** must be greater than the weight of the ship and its cargo.*

★ So what's going on?

Archimedes' Principle: For all objects that float, the mass of the liquid displaced is equal to the mass of the object. For all objects that sink, the mass of the liquid displaced is less than the mass of the object.

Archimedes was a mathematician and engineer who lived around 250 BC. While taking a bath he discovered the principle of buoyancy – how things float. The volume of water displaced was equal to the volume of his body. He got so excited when he noticed that the water level in the bath rose when he got in that he jumped back out yelling 'Eureka Eureka!'

Newton's Third Law: For every action there is an equal and opposite reaction.

A floating object, even you, will push a volume of fluid (which has a certain mass and weighs a certain amount) out of its way. This displaced fluid then reacts by pushing back on the object with the same amount of force that pushed it away. The force of

gravity that is pulling the object downwards is being counteracted by an upward buoyant force called upthrust. For a floating object, this buoyant force is equal to the weight of the displaced fluid (the fluid that is pushed out of the way by the object). Forces are measured in newtons.

Sink or float? That is the question!

If the density of an object is less than water then the object will sink into the water only to the point where the weight of displaced water is equal to the weight of the object.

SITUATION	SUMMARY
- Objects sitting on the top of the surface and held up by surface tension, e.g. leaves - Objects partly immersed in a liquid, e.g. ships - Objects wholly immersed but freely suspended, e.g. fish	Floating most frequently refers to objects partly immersed in a liquid but not actually touching the bottom. For example, icebergs float.
- Objects that are lower in the water than normal, e.g. ships in distress - Objects dropping through the water like a stone	Sinking most frequently refers to an object that is actually moving downwards through the liquid because the liquid cannot support it.

The captain's calculations: how much cargo can a ship carry?

Volume can be measured by submerging the object into water and measuring the volume of water it displaces: 1 millilitre = 1 cubic centimetre = 1 gram. The mass (in grams) of this displaced volume of water (in mL) equals the buoyant force of the water. This is the maximum mass that the ship and its cargo can have. Mass can also be calculated by weighing the object. Density can be calculated by dividing the mass of the substance by its volume.

The volume of a ship can be calculated by multiplying the area of the bottom of the ship by the depth of the ship below the water line. The weight that can be floated (ship + cargo) is equal to the weight of water that is displaced. This flotation capacity can be obtained by multiplying the ship's volume by the density of the water. The Plimsoll line is the line marked on a ship's hull indicating the maximum safe level to which the ship can be loaded in all types of seas.

The last word: Who is denser, girls or boys?

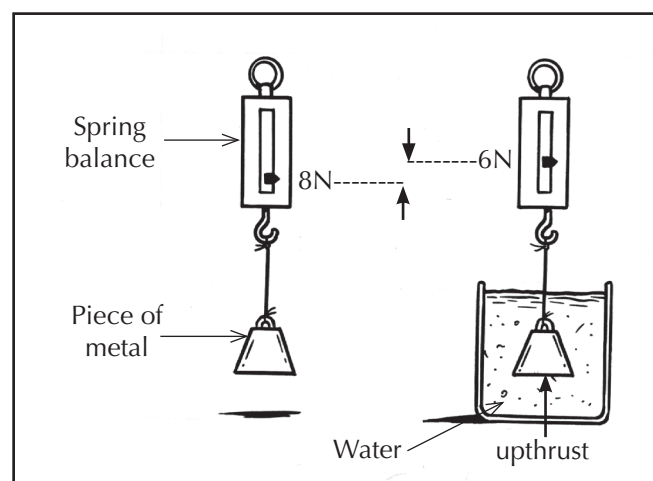
★ Figure it out!

- 1 a A group of students put a set of scales on a large tray. They carefully filled a jar to the brim with water, stood it on the scales and measured its weight. Then they dropped a small floating object into the jar and observed that some of the water overflowed into the tray.

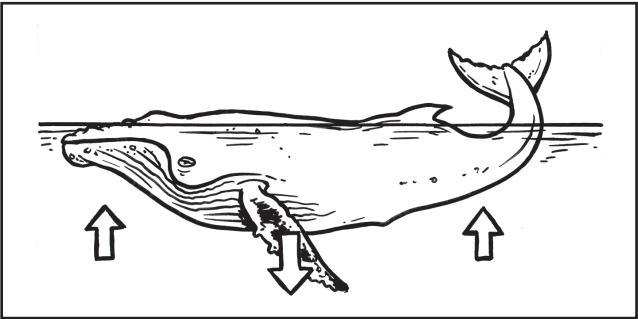
They measured the weight of this water and found that it weighed the same as the floating object. The students explained their results but left out some key words. Complete their explanation by filling in the missing words.

A floating object _____ a volume of liquid. The scientist _____ showed that the _____ of this water equals the _____ force exerted by the water on the floating object. This buoyancy, or upward force, is called _____.

- b Study the diagram above right. What is the buoyant force, in newtons, on the piece of metal?



2 Study the diagram of a whale basking in the sun. Using Newton’s Third Law, explain the forces that are acting on the whale and allowing it to enjoy this activity.



3 Ping-pong balls insist on floating on water. If you push them down in a glass of water they just pop straight back up.

a Why do they insist on floating?

b Explain this ‘popping up’ problem in terms of what you understand about buoyancy.

c What do you have to do to make them stay down?

4 If an object is less dense than water, it will float; if it is more dense, it will sink.

a How does a steel ship float, when steel has a density approximately eight times more than that of water?

b Rank the following shipping locations from 1 to 4, starting with the most dense and finishing with the least dense: European river, Antarctic seas, tropical seas, tropical river.

c The captain of a ship takes on cargo in Singapore in summer ready to travel to the North Atlantic. The Plimsoll line indicates that his ship is slightly overloaded. What will happen when he gets to the North Atlantic?

5 Why does an iceberg float?

6 Students are doing an experiment to investigate the density of various liquids. They rated the densities of the liquids from 1, which is the least dense, to 6, the most dense. Then, in a clear glass container, they floated each liquid one on top of the other according to their densities.

SUBSTANCE	fresh water	syrup	surgical spirit	salt water	olive oil	glycerine
DENSITY	3	6	1	4	2	5

a Which liquid is at the bottom and which one is at the top of the container?

b If an object had a density of 5.5, in which substance would it float?

c When ships sink, they usually leak a lot of oil into the ocean. Why is this dangerous to bird life and marine mammals?

7 Use Archimedes’ Principle to explain how submarines work.

8 In ancient Greece, King Hiero had a wonderful golden crown, but he was suspicious that it was not all gold, and could be a mixture of gold and silver. The king asked Archimedes to find out. Archimedes knew that gold is heavier than silver and that a crown made of a mixture of both would be lighter than one made only of gold.

How could Archimedes work out what the king’s crown was made of?

9 Consider a dinghy loaded with gravel and floating in a small dam.

a If the gravel is thrown overboard into the dam, will the water level in the dam rise, fall, or remain unchanged?

b What will happen to the water line on the boat? Will it be higher up on the dinghy, lower on the dinghy, or the same level on the dinghy?

10 Extend your understanding

a How could you work out the density of a block of wood that floats in water without knowing its mass?

b How could you work out the density of an object that sinks without measuring the volume of the object?

★ Give it the flick!

These experiments support Newton's First Law of Motion. The inertia of the coin keeps it in the same place even when the card is flicked out. Once the card is gone, the forces are not balanced and gravity pulls the coin straight into the glass. If the outside force acts quickly enough then the inertia of the object keeps it

from moving sideways. Inertia also keeps the stack of coins in the same place. When one coin is knocked out, gravity is then able to pull the rest of the stack down as far as the table, and so on with every coin that is removed.

★ The challenge

Embroidery hoops are inexpensive and can be purchased from haberdashery shops. The best bottles are 2 L rectangular-based fruit juice bottles with a neck diameter of about 3 cm (Just Juice brand is fine). The best 'hitting' technique will probably be to face the hoop toward you so that you can see the full circle. Swing your arm across your body horizontally,

quickly grabbing the far inside edge of the hoop while your arm is moving, and flicking your wrist to move the hoop from beneath the sugar cube. The sugar cube should drop into the bottle.

LINKS TO THE EVERYDAY – magic tricks, car accidents, racing cars around corners, passengers on a bus, moving anything

★ Curiosity trap

Use a piece of slippery polyester with no hemmed edges as a tablecloth and spread it onto a small, smooth table or desk. On top, place a heavy plastic dinner plate and mug. Grip the tablecloth firmly and pull quickly – almost snatch the tablecloth away horizontally in one fast, fluid motion. The dishes

hardly move as the cloth slips out from under them. If you used a long tablecloth or pulled slowly, the frictional force between the plates and the material would cause the plates to move. This is not magic, only science – the plates are heavy enough and have enough inertia that they don't start moving.

★ Answers

- 1 **a** the elephant **b** the elephant **c** the elephant
- 2 **a** 'A' – If you push a large mass in space, you feel the same amount of inertia as on Earth **b** 'A' is easier to stop, because it has less mass **c** The number of molecules in an object (its mass) does not change
- 3 force, inertia, same, seatbelts
- 4 **a** It should be a rapid and horizontal movement
b This is not magic, only science – the plates are heavy enough and have enough inertia that they don't start moving when the tablecloth does
- 5 **a** At first, both the seats and the children are stationary. When the bus starts moving, the seats also start moving – being attached to the bus. There is no new force acting on the children, so the children remain stationary – until the seats hit them in the back. Now both the seats and the children move in the same direction at the same speed **b** The children and the seats are both moving forward with the motion of the bus. When the brakes are applied, the seats stop moving as they are bolted to the bus; but since there is no new force acting on the children, the children continue to move in the same direction at the same speed, so they appear to be flung forward **c** The children keep going in the same direction at the same speed, but the bus and its attached seat go around the corner. The side of the bus pushes the child around the corner

- 6 **a** gravity and air resistance **b** It would travel forever at the same speed and in the same direction, as no force would be operating
- 7 A spinning object like a bicycle has its own momentum or inertia and resists change of direction, being moved or even stopped. A bicycle wheel demonstrates centripetal force and operates as a gyroscope
- 8 **a** The hoop moves, but inertia keeps the sugar cube in place. As long as the hoop does not move, the forces are equal. When the hoop is snatched away, gravity pulls the cube to the bottom of the bottle, where it again comes to rest **b** A stack of sugar cubes behaves the same way as a single sugar cube
- 9 **a** The cork moves in a circle **b** Your hand in the middle of the circle is exerting a force towards the centre **c** It will fly off in a straight line, as the force holding it in an orbit is no longer operating. **d** The enormous gravitational pull of the Sun holds the planets in orbit
- 10 **a** The inside and outside of a boiled egg all spin together. The raw egg has liquids inside, so the outside spins rapidly but the inside spins very slowly. The liquids inside exert friction on the inside of the egg, slowing it down **b** A cooked egg will sit still but a raw egg will begin to spin again as the fluids inside are still moving **c** Moving objects tend to stay in motion unless they are acted upon by some outside force

★ Give us a clue – give it the flick!

What you need

Drinking glass, pencil, a loop of cardboard 10 cm in diameter, a 10-cent coin, 1 playing card from the pack, 10 × 20-cent coins, a thin flat plastic ruler

What to do

Experiment 1

- 1 **Position** the card on top of the drinking glass. **Put** the 10-cent coin in the middle of the card.
- 2 Give the card a quick horizontal **flick** with your index finger. The card will fly out and the coin should drop into the glass. Try this again to perfect it.
- 3 Now **put** the cardboard loop upright in the drinking glass. **Balance** the 10-cent coin on top of the loop.

SAFETY!

- Make sure the coins do not hit anyone or anything in the room.

- 4 **Hit** the loop with the pencil so that the coin falls into the glass.

Experiment 2

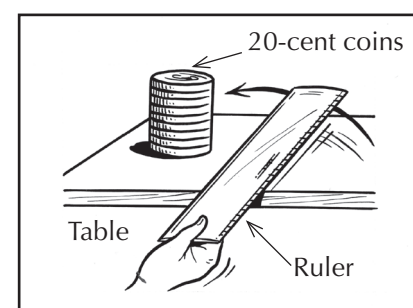
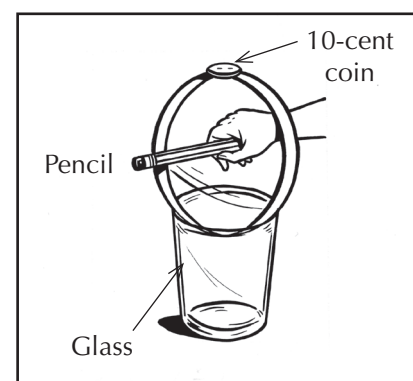
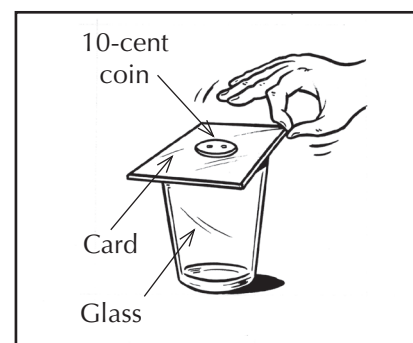
- 1 **Stack** the 20-cent coins one on top of the other in a neat pile on the table.
- 2 Firmly and quickly **sweep** the ruler along the table to knock the bottom coin out from under the pile. You may need to practise this.
- 3 **Continue** to sweep the ruler back and forth, knocking out the next coin and then the next until there is no stack left.

Now answer these questions:

- What keeps the 10-cent coin from moving sideways when you flick the card out?
- In both cases, what makes the 10-cent coin fall straight into the drinking glass?
- What keeps the stack of 20-cent coins in its original position when one of them is knocked out?

What happens next?

After you have completed these experiments and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!



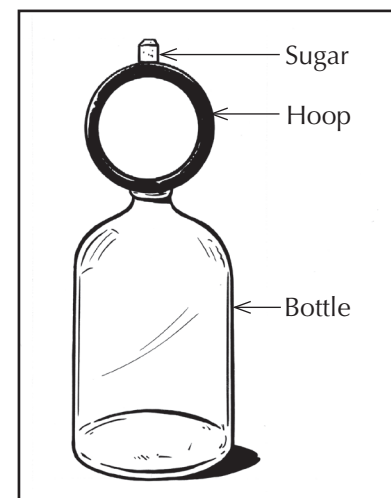
Clues!

Think of **matter as 'lazy'** – a mass that is resting will not move unless you give it a shove, and a mass that is coasting along will not alter its course unless you push or pull it. This is called **inertia**.

The more **sudden the movement** of the card, the more successfully you can do these tricks.

★ The challenge

Get as many sugar cubes as possible into the bottle by hitting the hoop with only one hand.



Materials

30 cm wooden embroidery hoop, 50 sugar cubes, a large, narrow-mouthed plastic bottle, a ruler

Instructions

You have 20 minutes.

- 1 Very carefully **balance** the embroidery hoop on the mouth of the bottle.
- 2 **Position** a sugar cube on top of the hoop as shown in the diagram below.
- 3 **Try** to get as many sugar cubes as possible into the bottle within the allowed time by hitting the hoop. You may use the ruler, but hold it in one hand only. (Tip: You may hold the bottle with the other hand so that it does not fall over.)

Clues!

Remember the stack of coins in 'Give it the flick?'

*The 'reluctance' of an object to start moving is called **inertia**.*

★ So what's going on?

What's inertia?

Inertia is a property possessed by all matter; it can be thought of as laziness or 'difficult-to-move-ness'. The tendency of matter to keep doing what it is already doing (even if it is doing nothing) is its inertia.

Every object in the universe has its own inertia and will not move unless something forces it to move. So if anything starts to move, you can be sure it has been pushed or pulled. There had to have been unbalanced forces operating.

Similarly, an object that is already moving will keep moving in the same direction and at the same speed, until a force makes it change. This form of inertia is called momentum – the force of a moving object.

Overcoming an object's inertia and getting it moving depends on its mass. Slowing an object down if it is already moving also depends on its mass. Mass is the amount of matter, or atoms, in an object; as mass increases, so does inertia. Since inertia depends on mass, the inertia of an object will be the same on Earth as it is in space.

Newton's First Law of Motion: An object at rest remains at rest, and an object in motion remains in motion, until acted on by an external force.

An object will only start to move if a force is applied to it, and once it is moving it will carry on at the same speed in the same direction. In other words, moving objects need a force to stop them and stopped objects need a force to get them going again.

Sir Isaac Newton's great realisation was that in the absence of equal forces, continuous straight line motion will occur. However, identifying all the forces acting on a moving object is very difficult – some are invisible. Examples of these '**hidden forces**' that act on moving objects are friction, gravity and buoyancy.

Circular motion

Forces are supposed to act in straight lines, so why do things like fans, discs and propellers move in circles? A spinning object is always changing direction. The force that continually pulls towards the circle's centre is called **centripetal force**. This force may be the result of the spokes of a bicycle wheel or it may be the force of gravity. If this force stops, there is nothing to pull the object around in a new direction and it will at once fly off in a straight line. Something moving in a straight line has momentum and resists changing direction. A spinning object also has its own momentum, called angular momentum. It too resists being moved or stopped.

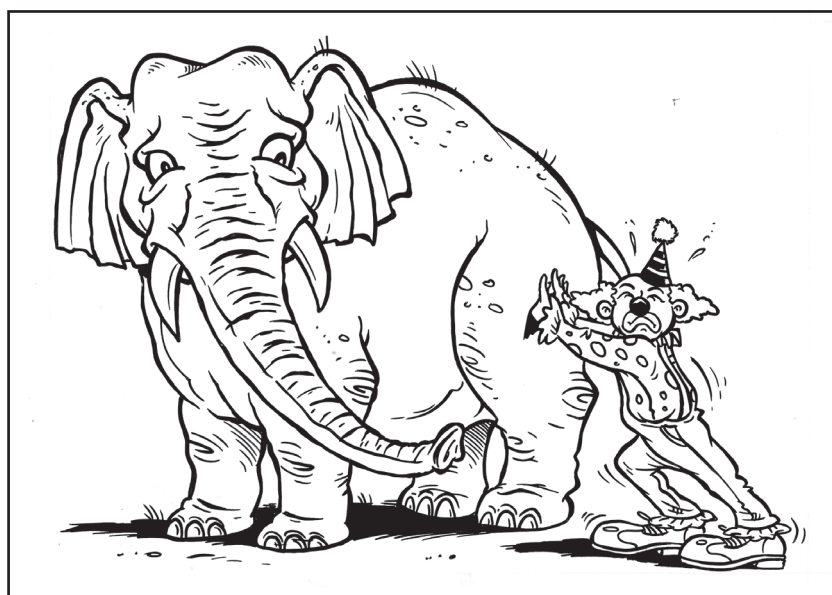
Spinning objects, like spinning tops and gyroscopes, resist changes to their plane of spin or axis of rotation. A gyroscope that is spinning fast enough has enough momentum to resist changes caused by gravity. Gyroscopes are so stable that they are used in navigation systems.

The last word

Do you have too much inertia to get out of bed and go to school some mornings?

★ Figure it out!

- 1 Look at the sketch of the clown and the elephant.



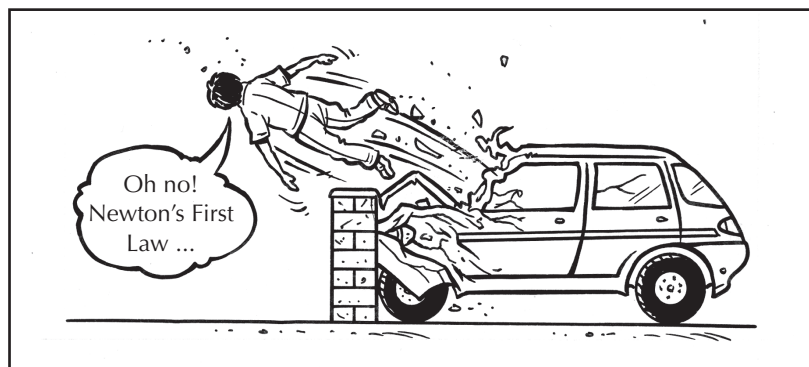
- a Who is the most difficult to move?
- b Who has the most mass?
- c Who has the most inertia?

- 2 Look at the two supermarket trolleys in the diagram.



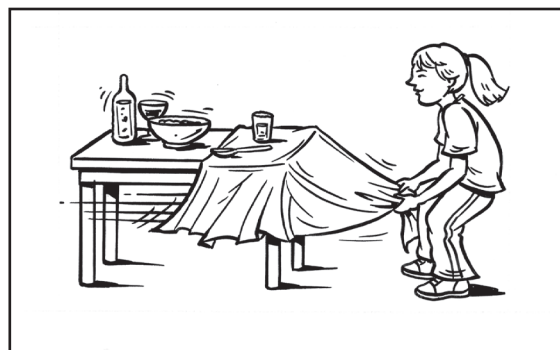
- If you pushed both trolleys with the same force, which one would be the easiest to get going? Why?
 - Which trolley is easier to stop? Why?
 - If you took both trolleys into space, would they weigh the same?
- 3 Car companies conduct crash tests on their cars so they can make them safer for passengers. Complete the following explanation.

Moving objects need a _____ to stop them as their _____ tends to keep them moving in the _____ direction, at the same speed. The force from the _____ that are fitted to cars must be big enough to slow the passengers down and minimise injury when the car hits a wall.



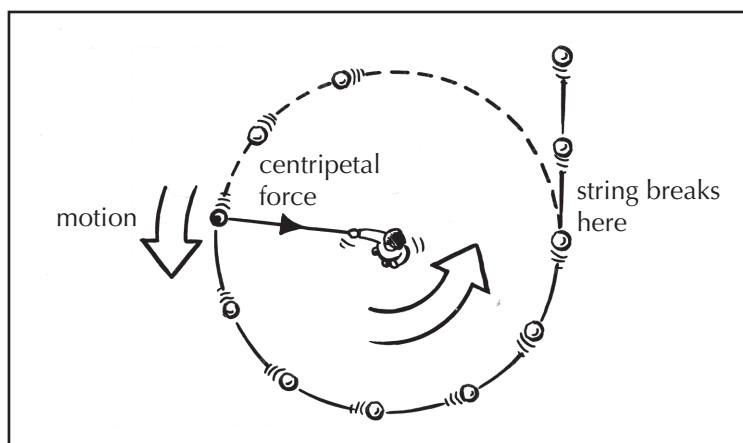
- 4 Your teacher presented a 'magic' trick at the beginning of this session. It was based on the idea that things like to keep on doing what they are already doing and that they do not like to start moving or stop moving – they are lazy!

The diagram shows the trick.



- Should this person pull the tablecloth quickly or slowly? Why?
- Explain the physics of the trick.

- 5 Children are travelling to school in a school bus. Explain the following situations.
 - a When the bus takes off quickly, they feel like they are thrown back in their seats.
 - b When the bus brakes suddenly, they feel they are thrown forward in their seats.
 - c Why do children hit the side of the bus when the driver takes a corner too fast?
- 6 A sportsperson is playing a game of golf. The ball is resting on the grass, has inertia and appears to be very 'lazy'. The sportsperson gets it moving by adding a force to it – hitting it very hard with a golf club.
 - a What hidden forces are now operating on the moving golf ball?
 - b If these hidden forces could all be eliminated, what would happen to the golf ball?
- 7 Riding a bicycle demonstrates Newton's Laws of motion in action. Why is it much easier to sit on your bike when it is moving along?
- 8 Refer back to the challenge.
 - a What techniques did the winner of the challenge use?
 - b Explain why the winner's technique was successful.
- 9 If you whirl a cork tied to a piece of string around your head, you can feel the force in the string. Newton's First Law of Motion says that the cork will continue to move in a straight line unless a force is applied.



- a In what direction does the cork move?
 - b What force is stopping the cork from going straight ahead?
 - c What will happen to the cork if the string breaks?
 - d What holds the planets in our solar system in their orbits?
- 10 Extend your understanding**
- A chef wanted to find out whether an egg was raw or boiled. She knew to spin them both on the table. The boiled egg spins easily, evenly and quickly. The raw egg is difficult to spin and only moves slowly.
- a Explain, in terms of science, how the chef solved the problem.
 - b What would happen if you used your finger to briefly stop the eggs spinning, then quickly pulled your finger away?
 - c Analyse this phenomenon in terms of Newton's First Law of Motion.

★ Tide's rising!

Experiment 1

While the candle burns, the air inside the jar gets hotter, takes up more space and forces some of the air out – bubbles emerge from under the jar. When the candle goes out, the smaller mass of air still trapped in the jar cools, the volume decreases and water rushes into the space formerly occupied by the air that escaped. This supports Boyle's Law (see 'So what's going on?').

★ Challenge

As the teacher you should demonstrate the safe use and disposal of matches, as well as the procedure for getting the egg into the bottle, before students begin. Using matches to get the egg into the bottle may be a safety risk with some students. If so, the teacher can light the newspaper once each team has their egg and bottle ready. Matches are no longer required once the egg is in the bottle and should be collected immediately. 'Spring Valley' fruit juice bottles and medium-sized eggs work well. If the egg breaks in half as it enters the bottle, use a smaller egg or a bottle with a slightly larger neck.

As an alternative, water balloons adjusted to the right size for whatever bottles are available can work well in place of eggs. However, the balloons will not break. The egg is pushed into the bottle by the stronger air

Experiment 2

Applying pressure to the sides of the bottle increases the pressure on the water, forcing water into the pen lid and reducing the volume of air trapped in the top of the lid. The pen lid thus becomes heavier and sinks. Releasing the pressure allows the volume to increase, forcing out some of the water. Since the lid now weighs less, it will float. This supports Charles' Law (see 'So what's going on?').

pressure outside the bottle.

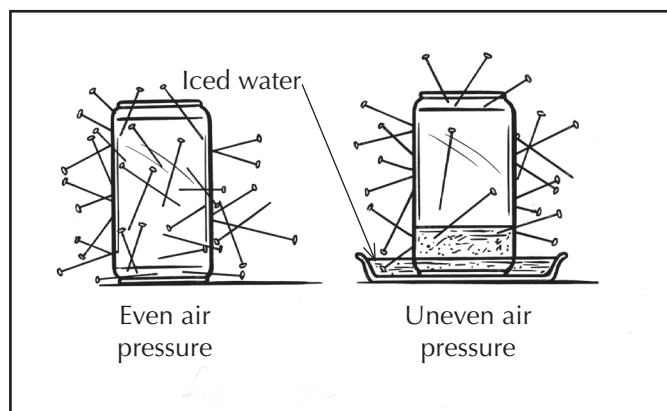
Applying this same basic principle allows the egg to be removed again. Either blow more air into the bottle, using a straw flattened between the egg and the glass, or increase the pressure of the remaining air by gripping the bottle with tongs and holding it under hot running water – but remember, these bottles are not pyrex and may crack. In both cases, the pointy end of the egg should be firmly positioned in the neck of the bottle to prevent any air particles from escaping. It also helps to rinse the ash out of the bottle before beginning.

LINKS TO THE EVERYDAY – bubble wrap, milking machines, pressure pumps, weather patterns, vacuum cleaners, hot-air balloons, car tyres.

★ Curiosity trap

Before you start, half-fill a bucket with water and crushed ice to make a very cold ice slurry. Put a very small amount of water (1–2 mL) into a soft drink can, grip the can with a pair of kitchen tongs, and heat the water in the can over a hotplate or Bunsen burner until water vapour is pouring out of the can. Now very quickly plunge the can, open end first, into the ice slurry. The can will totally collapse with a loud and satisfying 'thwack'. If this doesn't work the first time, try again, making sure that the water in the can has almost all vaporised and that the top of the can is completely submerged in the ice.

If you seal hot air inside a container and gradually allow it to cool, the air particles inside will slowly lose energy, move closer together and take up less space. Thus the pressure inside the container drops relative to the normal atmospheric pressure outside. Because there is less air inside the bottle than there was to start with (a partial vacuum) the pressure inside the bottle



is a lot lower than the outside pressure of the air. The cooler the trapped air becomes, the more crushed the can will become.

This is an example of Charles' Law – as the temperature of air decreases, its volume also decreases.

★ Answers

- 1 Temperature is directly proportional to volume and if the temperature rises, then the molecules have more energy and move faster, taking up more space. The converse is also true
- 2 volume, temperature, force, more often
- 3 Your middle ear is separated from your outer ear by a flexible membrane. The higher you go, the less the pressure there is on the outside of this membrane. The pressure inside remains at ground pressure, which is higher. When the pressures equalise, your ears 'pop'
- 4 **a** The air in the flask is pushing upwards on the water in the funnel with the same force as the atmosphere is pushing downwards. The forces are balanced, so nothing happens (Newton's Third Law) **b** The water stays in the glass because the pressure of the outside air against the cardboard is greater than the pressure of the water against the cardboard – unbalanced forces
- 5 **a** clear skies and fine weather **b** further apart **c** wind **d** Winds are very strong because the high pressure and low pressure areas are so close and the pressure difference is so large that the air moves very, very fast
- 6 **a** As the temperature rises, volume goes up; and as the temperature goes down, so does the balloon's volume **b** Charles' Law **c** Boyle's Law **d** They move faster and further apart
- 7 **a** The balloon will inflate **b** The air particles in the flask will be pumped out, so the air pressure inside the flask will be zero (a vacuum) and atmospheric pressure will therefore be able to inflate the balloon **c** The balloon will deflate **d** The motor of the vacuum cleaner creates a partial vacuum, so air rushes in and takes dirt and dust with it
- 8 The pockets of air are trapped inside plastic film. When they walked on it the air was forced to take up less space, but their weight was spread over a large area. When they used their heels the air was forced into a much smaller volume and the plastic film was not strong enough to withstand the increased pressure – so it failed and went 'pop'. This is explained by Boyle's Law
- 9 **a** As the air is heated, the molecules take up more space and the air becomes less dense and therefore lighter than the air around it – hot air rises **b** The greater the difference in densities between the inside (very hot) and outside air (very cold), the greater the lifting power **c** The air is either cooled down slowly or some of the hot air is let out through the vent so it can be replaced with cold outside air
- 10 **a** When you drink through a straw, you lower the pressure in the straw; atmospheric pressure, pushing on the surface of the liquid, forces it up the empty straw and into your mouth. A hole in the straw allows air to be constantly drawn into the straw, so the pressure cannot be lowered and liquid cannot be forced into the straw – there is no space **b** He could solve this problem by covering the hole with his finger. Turning the straw up the other way so that the hole was covered by the liquid would work while the depth of liquid in the cup remained more than 3 cm

★ Give us a clue – tide's rising!

What you need

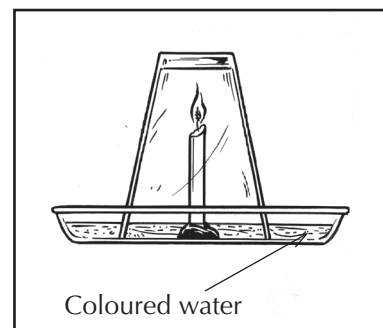
Experiment 1: birthday candle, a little plasticine, glass jar or beaker, shallow dish, coloured water, matches

Experiment 2: 2 L soft drink bottle, pen lid, plasticine, water

What to do

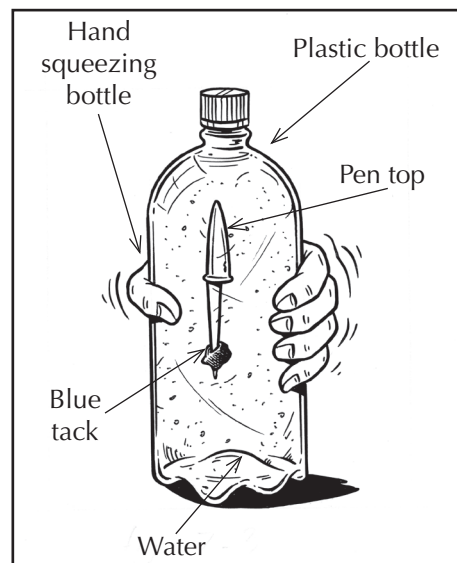
Experiment 1

- 1 **Stick** the candle to the bottom of the dish using the plasticine.
- 2 **Half fill** the dish with coloured water.
- 3 **Light** the candle.
- 4 Quickly **invert** the jar over the candle.
- 5 **Watch** very carefully and observe what happens. You may need to do this experiment several times in order to see everything that goes on!



Experiment 2

- 1 **Remove** the label from the bottle. Fill the bottle to the very top with water, leaving no air space at all.
- 2 **Add** a small weight of plasticine to the arm of the pen lid so that it will float in a container of water with just the tip of the lid above the surface. Adjust the amount of plasticine until you achieve this. Make sure there is no hole at the apex of the lid, or seal it with a tiny bit of plasticine.
- 3 **Float** the pen lid in the neck of the bottle and screw on the lid.
- 4 **Squeeze** the sides of the bottle firmly. The pen lid will sink. If it does not sink, add a little more plasticine. When you release your grip, it will rise.



Now answer these questions:

- In Experiment 1, why did the water rush into the jar when the candle went out?
- Explain what happened in Experiment 2 in terms of the relationship between the pressure and volume of the air trapped in the pen lid.

What happens next?

After you have completed Experiments 1 and 2, and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from these experiments and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

*The more a fixed volume of air is heated, the **more pressure** it has.*

*Pressure changes travel unaltered through water but are rapidly absorbed by air. If a fixed volume of gas is forced into a smaller container then the **pressure will increase**.*

*If molecules in a container are given more energy, they move faster, hit the walls harder and more often and take up more space, so the **pressure rises**.*

*Forcing more air particles into the same space will also cause an **increase in pressure**.*

★ The challenge

Devise a plan to remove the egg from the bottle without breaking the bottle or damaging the egg. There are several ways to achieve this.

Materials

A straw, a pair of tongs, 1 hard-boiled egg, matches, paper towel, a strip of newspaper 12 cm × 1.5 cm, a small glass bottle with an opening large enough to let an egg almost pass through, water, access to hot water

★ Instructions

- 1 **Peel** the egg carefully so that there are no chunks missing and the egg is smooth on all sides, except for the flat area at the top where the air sac was. If this flat area is on the side of your egg, ask for another.
- 2 **Practise** sitting the 'pointy' end of the egg in the mouth of the bottle. The egg should be just a little bigger than the opening. Remove the egg.
- 3 **Wash** the egg in water and leave it wet.
- 4 **Carry** this step out smoothly and as quickly as possible. Hold one end of the newspaper, light the other end, and calmly drop the burning paper into the bottle. Carefully and firmly position the egg in the mouth of the bottle as you did above.
- 5 **Watch** very carefully what happens!

SAFETY!

- ☛ You will be using matches. Make sure you strike and handle them carefully.
- ☛ Your teacher will demonstrate the correct way to use and dispose of matches, as well as how to safely ignite the newspaper. Watch carefully.
- ☛ Once the egg is in the bottle you will no longer need the matches, so return them to the teacher.

What really happened?

As the air in the bottle was heated, the molecules began to move faster and further apart, so some of them escaped through the mouth of the bottle. The egg sealed the mouth of the bottle just before the flame went out, so no more air could get into or out of the bottle as the air inside cooled. The number of molecules remaining in the bottle was now lower, so the air pressure was lower than normal atmospheric pressure. The greater pressure outside the bottle pushed the egg into the bottle. There was space for the egg because some of the air molecules had escaped.

- 6 Using only the materials provided, work out a way to get the egg out of the bottle without breaking the bottle or damaging the egg.

Clues!

*The egg can be removed from the bottle by applying the **same principle** that put it in there in the first place.*

Air occupies space, which means that nothing else can occupy the space at the same time.

☆ So what's going on?

The atmosphere

We live under an ocean of air about 100 km thick called the atmosphere. It surrounds the planet and is made up of a number of different types of gas molecules, such as oxygen, nitrogen, water vapour, hydrogen etc. These molecules, or particles of matter, all weigh something and take up space, so they tend to push on every other molecule and object they collide with. The force of these collisions is equal to about 1 kg per square centimetre. This is called air pressure. Because air at higher altitudes compresses the air below it, about half of the total weight of the atmosphere is concentrated in the 5 km closest to the Earth. The force of gravity keeps the molecules that make up our atmosphere close to the Earth.

What is air pressure?

Air pressure makes things happen. It can be extremely powerful, but its effect is only obvious if the air pressure is altered from its natural state. Air pressure works equally in all directions and on all exposed surfaces; it flows into spaces not already occupied by air.

When air particles heat up, or absorb energy, they move faster and further apart, so a given volume of air will take up more space. In a 'closed' system, the pressure of the air particles on the inside of the container will increase. In an 'open' system, the air will move from areas of high pressure to areas of low pressure.

Air pressure is strongest near the surface of the Earth. It decreases as you go higher into the atmosphere.

The atmosphere and the weather

The atmosphere is responsible for all our weather patterns. In high pressure areas (H) the air molecules are closer together and produce clear skies and fine weather. Low pressure systems (L), where the particles are further apart, tend to bring rain and storms.

Some weather patterns occur because two masses of air with different temperatures and pressures are vying for the same space in the atmosphere. Moving air is called wind. Air moves from areas of high pressure to areas of low pressure. The larger the pressure differences between these areas, the stronger the wind.

Boyle's Law: The volume of a gas is inversely proportional to its pressure, if all other variables are held constant.

If the volume of a fixed mass of gas is reduced, the molecules will be closer together. For example, if a fixed amount of gas is forced to take up half the volume it originally took up, its pressure will double.

Charles' Law: Temperature is directly proportional to volume, assuming that all other variables remain constant.

In other words, if the molecules in a certain volume of gas start to move faster (for example through heating), they will move further apart. If we want the pressure to stay the same, they have to take up more space.

The Pressure Law: If the temperature rises, molecules have more energy and move faster. If the volume of a container is kept constant, the molecules will hit the walls harder and more often, so the pressure will rise.

★ Figure it out!

1 A group of students pulled a new balloon over the neck of a small glass bottle and then put the bottle into boiling water. They observed that the balloon got a little bigger. When they put the bottle into iced water the balloon deflated. Explain these results by applying Charles' Law and the Pressure Law.

2 Fill in the words that are missing from the information that follows.

You can increase the pressure of a gas:

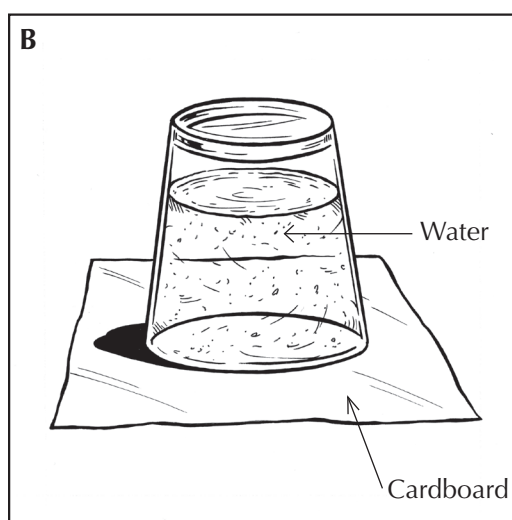
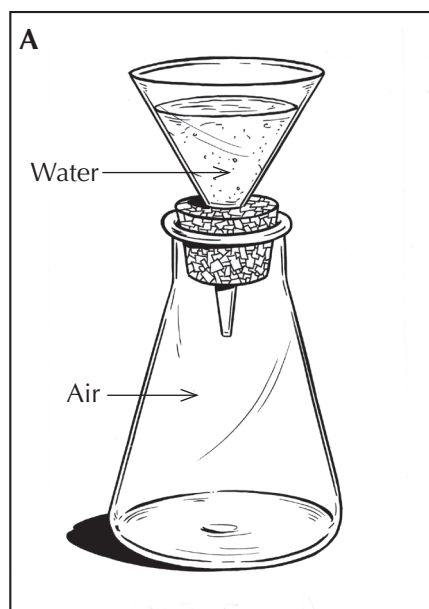
- By squeezing it into a smaller _____.
- By increasing its _____, which makes its particles hit the sides of the container with more _____.

Both squeezing and heating the gas make its particles hit the sides of the container _____.

3 Why do your ears 'pop' when you drive up a mountain or fly in an aeroplane?

4 a Study diagram A. Why won't the water go into the flask?

b Study diagram B. Why doesn't the water in the glass push the cardboard off?



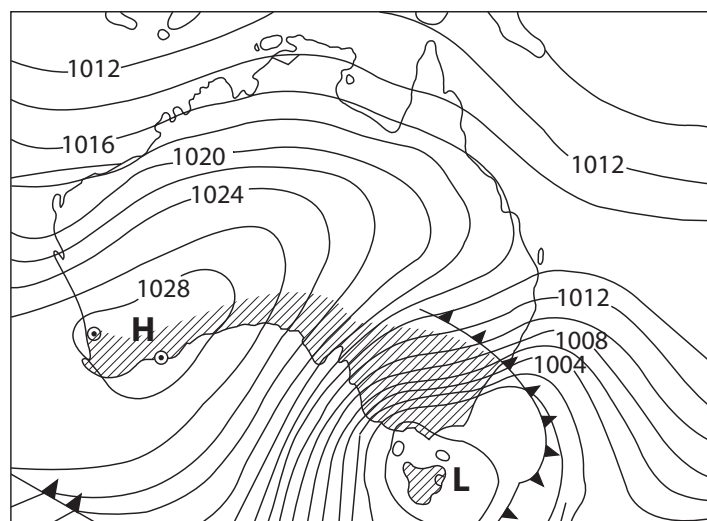
5 Study the weather chart here and answer the questions.

a What will the weather be like in the area marked H?

b Will the molecules in the atmosphere be closer together or further apart in the area marked L?

c What term do we use to describe air particles moving from a high pressure system to a low pressure system?

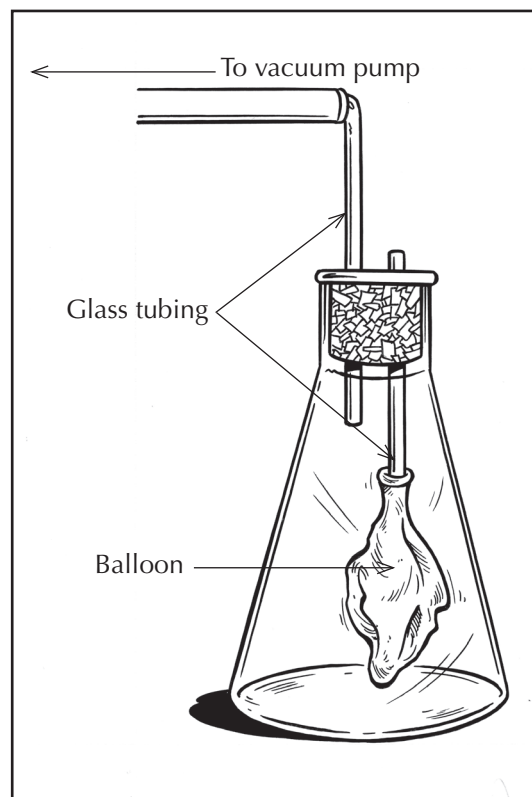
d A cyclone is a high pressure system surrounding an 'eye' of very low pressure. Why are the winds so extreme in a cyclone?



- 6 Students carried out an experiment to investigate air pressure. They half inflated a balloon, tied it off, then measured its diameter. After leaving it in a freezer for 10 minutes, they measured it again. Then they put it into a bucket of very hot water for 10 minutes and recorded the diameter in their data table. Study the data and answer the questions that follow.

	AT ROOM TEMPERATURE	AFTER 10 MINUTES IN THE FREEZER	AFTER 10 MINUTES IN BOILING WATER
Balloon diameter (cm)	10 cm	8 cm	12 cm

- a What appears to be the relationship between the volume of the balloon and the temperature?
- b Which gas law do these results support?
- c Which gas law does the Challenge support?
- d What two things happen to particles of matter when they are heated?
- 7 A vacuum pump will pump air from an enclosed system. The diagram here is an example of such a system.



- a When the pump is turned on, what will happen to the balloon?
- b Explain your answer to (a) in terms of air pressure.
- c What will happen to the balloon when the pump is turned off?
- d How does a vacuum cleaner work?
- 8 Some children were playing with bubble wrap that arrived as packaging material. They had fun walking on it in bare feet, but it wasn't until they started to walk only on their heels that the wonderful popping sounds began. Explain why the bubbles only burst when they used their heels.
- 9 Hot-air ballooning originated in France. Balloons need a good burner and a lightweight material that is airtight and resists heating. A venting system in the top of the balloons releases hot air. To fly the balloon, the air inside is heated by the burner.
- a Why does the balloon rise into the air?
- b How could the lifting power of the balloon be improved?
- c How is the balloon returned to Earth safely?

10 Extend your understanding

Some friends at a party were drinking soft drink using paper cups and straws. One had a damaged straw – it had a hole about 3 cm from the end he was trying to drink from. He was having great difficulty getting any of the drink into his mouth.

- a Why was he unsuccessful at quenching his thirst?
- b How could he fix the problem using the same straw?

The last word

How long can a straw be and still allow you to drink a whole can of soft drink?

★ Marshmallow sky scraper

Encourage students to experiment with ways to join the marshmallows together with toothpicks. Is there any advantage in having more than one toothpick coming out of one marshmallow? Ask students to predict how

high their tower will be. As a material, marshmallows are full of air and not very dense; they are squishy and cannot support much weight. Toothpicks, on the other hand, are quite rigid.

★ The challenge

All groups should use books weighing the same (class sets of textbooks are good). Alternative materials such as drinking straws could be used to build the structure.

LINKS TO THE EVERYDAY – bridges, water towers, buildings, trees, animal skeletons, honeycomb, spider webs, aircraft, cranes, tin cans and milk crates, dams, tunnels.

★ Curiosity trap

Take one lightweight (disposable) cup and two drinking glasses. Turn the two glasses upside down and place a piece of paper the size of an envelope on the bases so that it spans the distance from one glass to the other. Half fill the lightweight cup with water and position it in the middle of the paper. The paper will not be rigid enough to support the cup.

Now pleat the paper lengthwise and repeat the experiment. The same material has now been strengthened enough to support the weight of the cup.

The downwards force (pull of gravity) on the cup is now spread out or dispersed along the length of the paper. It is acting on a larger surface area.

★ Answers

- 1 **a** Gravitational force downwards, upward force exerted by the structure **b** Strong structures transfer the energy to the table, where it spreads out over a larger surface area **c** The books would fall and break the structure
- 2 He must push up with the same amount of force as the weight of the dumbbell
- 3 A larger cross-section will always be stronger, because there is more area to share the force
- 4 If a mass of 100 g = 1 newton then a weight of 50 000 g = 500 newtons
- 5 **a** A **b** B – because compression at the apex was resisted by tension in the string **c** compression at the apex and tension in the string
- 6 Examples are bee hives, spider webs, coral reefs, birds' nests, the buttress roots and aerial roots of rainforest trees, skeletons of large mammals like elephants
- 7 **a** use of triangles, overall pyramid shape, several pieces of fettuccini taped together to make beams, structured to resist tensile forces, the structure spreads the load via the beams onto the table **b** the tape between the beams – called a strut **c** to prevent bending
- 8 **a** Compression force – the weight and load on an arch bridge push downwards, but the arch turns this force along the curve of the arch itself and into the strong supports on the banks **b** Because very large ships must pass under the bridge to get into the harbour
- 9 Newton's Second and Third Laws combined say that the force of the book on the spaghetti structure equals the weight of the book (gravitational force acting on the book). The book falls when the downward force breaks the structure and the book accelerates towards the centre of the Earth. When the book strikes something, such as the floor, that can exert enough upward force to balance the book's downward force, it comes to rest
- 10 When you change the shape of paper it becomes much harder to crush. Cut or fold the newspaper and roll it into cylinders 10 cm long. Hold the paper bundles together with elastic bands and set them upright under the wood. Twelve to 15 of these cylinders will support the weight of the average teacher

★ Give us a clue – marshmallow sky scraper

What you need

50 toothpicks, 20 ordinary marshmallows, ruler

What to do

- 1 **Plan** to build the tallest tower you can, using the materials provided. To be considered a success, the tower must stand long enough to be measured.
- 2 **Predict** how high the structure will be and jot your estimate down.
- 3 Before you begin to build, **consider** how both of the materials behave, how you might join them together, and how you might strengthen them.
- 4 If your first design fails, either **modify** it or try a completely different idea.

Now answer these questions:

- When a structure fails, you learn from it. What information have you discovered from building this structure that will help you design a different structure?
- Which shapes appear to be the strongest?
- What is it about the marshmallows that limits their use as a building material?
- Which group built the tallest marshmallow tower? Why was it so successful?
- Comment on the joins between the marshmallows and the toothpicks.
- How did the final height of the tower compare to your predicted height?

What happens next?

After you have completed this experiment and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

*Different **materials** behave differently. Their different properties influence and also limit the way the materials are used.*

*Different **shapes** of the same materials have different strengths.*

Materials can be strengthened and weakened.

*Methods and materials used to **join** the parts of structures must provide strong but flexible connections.*

★ The challenge

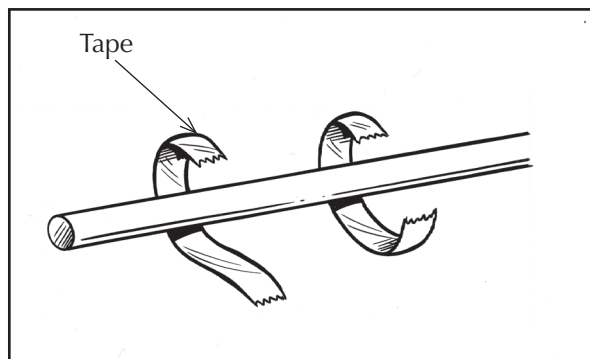
Design and build a structure, or group of structures, that will exert an upward force on a given weight equal to the downward force exerted on the weight by the Earth.

Materials

20 pieces of undamaged, uncooked spaghetti, ruler, 1 m of masking tape, a heavy book, scissors, stopwatch

Instructions

- 1 Using 20 pieces of spaghetti and 1 m of masking tape, build a structure or structures that will support the weight of at least two of the books provided for at least 30 seconds and at a minimum height above the table of 5 cm.



Taping several pieces of spaghetti together forms a stronger beam.

- 2 You have 20 minutes to complete the challenge and then the judge will test the structure using two books.
- 3 If the structure supports two books then three books will be used.
- 4 You do not have to use all the pasta or masking tape in building this structure.

SAFETY!

- ⚠️ Wear safety goggles or other eye protection when placing books on your structure as the pasta could shatter and produce sharp fragments.

Clues!

The structure must provide a force equal and opposite to the force exerted by gravity on a stack of books.

If the squashing or stretching force on a structure is shared or spread out, it can resist more force. Comparing one material with another tells us how much force a material is able to resist.

Pasta resists compression forces very well, but it breaks easily when twisted or bent.

In your design you should plan to use as much of the strength of each piece of spaghetti as possible and ensure that the individual pieces work together to provide an overall strength.

☆ So what's going on?

What is a structure?

All structures have to:

- hold up or support something (e.g. a building)
- reach across or span a distance (e.g. a bridge), or
- contain or protect something (e.g. an aluminium can).

Structures must support at least their own weight and often much more. Different forces balance each other to form stable structures. Strong, stable structures are able to transfer the weight of the load to another place as well as spread out the force.

We say that a structure is loaded when a force acts on it. When a structure is put under load it must be able to resist it. A structure fails when one part breaks, fractures, buckles, bends, twists or stretches and is permanently distorted.

Properties required of structures

Structures must be made of strong materials and designed to give maximum support.

- Strength is how much force a material can stand before it breaks. In other words, its ability to resist forces that push together (compression) and forces that pull apart (tension) tells us how strong a material is. Compressive forces (squashing) cause buckling and bending of materials. The part of the structure that resists these forces is under tension (stretched) and is called a strut. Materials must be able to be compressed and stretched without permanent distortion (change of shape).
- Flexible materials give a little when under load, but inflexible ones may snap instead of bending.
- Rigidity is the ability to withstand being bent, twisted, stretched or changed in shape.

The shape of things to come!

- Triangles are the most rigid shapes because they resist twisting. A triangle is also a very stable shape and is the key to building frames which are both strong and stable. When a force is exerted on a triangular shape, all the sides work together to hold it in place. If the triangle is made of a strong material, it will resist enormous loads.
- Tubular shapes are strong and also resist twisting and bending. Bundles of tubes are stronger than just one. A tube is much lighter in weight than a solid cylinder.
- If a structure and its load are balanced then the forces operating must be equal and opposite. Stable structures exert an upwards force on an object that is equal to the downward gravitational force. When forces are equal and opposite, the structure is stable, so there is no change in motion (it does not collapse).
- According to **Newton's Second Law** of motion, the net force acting on the book must be zero; and since the book is not moving, its acceleration is also zero. The upward force exerted by the spaghetti structure supporting the book is equal and opposite in direction to the weight of the book. The two forces involved that are described by **Newton's Third Law** are the downward force the book exerts on the structure and the upward force the structure exerts on the book. These forces are equal in magnitude and opposite in direction. If these two forces are equal, the book will be supported and the structure will not distort.

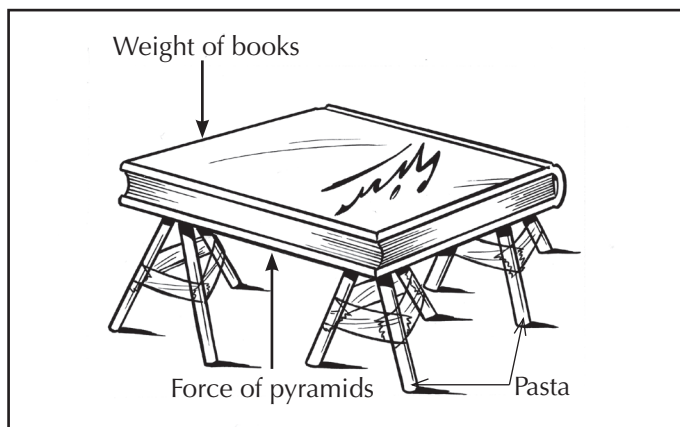
What has Newton got to say?

Newton's Second Law: The greater the force on an object, the greater its change in speed or direction; and the heavier an object, the less its change in speed or direction.

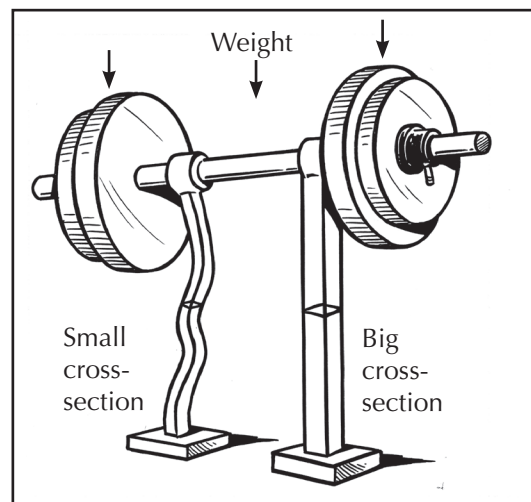
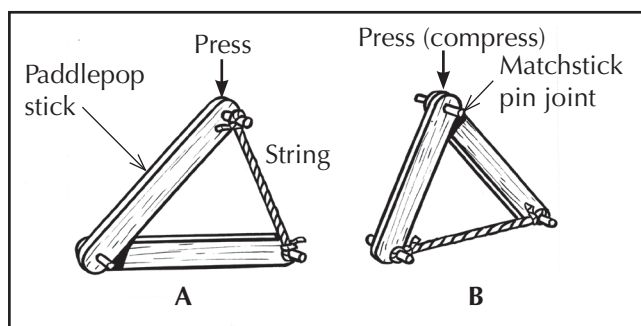
Newton's Third Law: For every action there is an equal and opposite reaction.

★ Figure it out!

- 1 Study the diagram below.

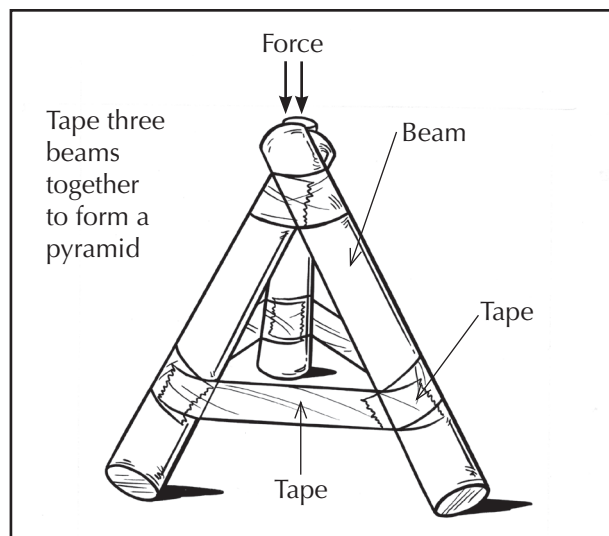


- a Name the equal but opposite forces acting on the book.
 - b In terms of energy transfer and dispersion, explain why this solution to the pasta challenge would work.
 - c What would happen if the upward force from the pasta structure were smaller than the downward gravitational force on the book?
- 2 When a structure is put under load, it must be able to resist it. When a weightlifter holds up a dumbbell, how much force must he use if he is not going to crumple under the load?
 - 3 A structure designed so that the energy is able to spread out allows the object to resist more force. Bearing this in mind, explain what is happening in the diagram.
 - 4 Engineers often do tests and calculations to find out how a structure will behave under load or what weight it can safely carry. If a shopping trolley is carrying a mass of 50 kg, what downward force in newtons is acting on it?
 - 5 To find out about how forces act on a triangular frame, a student made up a triangle with paddle-pop sticks, matchsticks and string. Study the diagram and answer the questions.



- a In which situation (A or B) did the triangle collapse?
- b In which situation did the triangle resist the force? Why?
- c Where are the compression and tension forces operating in situation B?

- 6 Animals and plants in the natural world construct some amazing structures. For example, the tallest trees in a rainforest and the intricate, high-rise nests of termites with their own air conditioning. List another five impressive natural structures.
- 7 The structure in the diagram has been made out of sticky tape and fettuccini.



- a List five design techniques that make this structure strong.
 - b What part of this structure is designed to resist tension? What term is used for this?
 - c Why was sticky tape wound around the beams of pasta?
- 8 The Sydney Harbour Bridge is a steel arch bridge. It is one of the world's heaviest and widest bridges. Steel arches are constructed by riveting or welding beams in place one at a time. The ends of the bridge are constructed first and are enormously strong. The arch is then built out from each end until it eventually meets in the middle. The bridge will not support itself until the arch is complete.
 - a Arch bridges are unique because their strength relies on only one force. Which force is it?
 - b Why did the Sydney Harbour Bridge need to be built so that it arched up over the water with no supports in the harbour floor?
 - 9 Using Newton's Second and Third Laws, explain why the book in the challenge might smash the spaghetti structure and come to rest on the table or the floor.
 - 10 **Extend your understanding**

As part of an examination, students were asked to use a newspaper, some elastic bands and a thin piece of wood to design a structure that would support a teacher of average weight, 10 cm above the level of the floor. Newspaper is thin; it buckles and folds under load and crumples easily. Wood and elastic bands behave quite differently. Work out a plan which will satisfy the examiner.

★ Gathering momentum

The basketball hits the table first and reverses its motion. The tennis ball collides with the basketball as it is moving upwards. So the change in momentum of the tennis ball is much larger than it should be. When the ping-pong ball hits the tennis ball, the very light ping-pong ball bounces off the tennis ball with

combined momentum and energy, so it bounces very high. The extra gravitational potential energy of the ping-pong ball is accounted for by the fact that the other two balls achieve heights slightly lower than they would have in the absence of the ping-pong ball. Energy and momentum are therefore conserved.

★ The challenge

Mess is not usually much of a problem but you may like to put a few sheets of newspaper in the 'drop zone' in case of an untidy landing. This is a collision between the Earth and the designed structure. The forces are equal and opposite (Newton's Third Law) and the changes in momentum are also equal and opposite. In all collisions, momentum as well as

total energy is conserved. A detailed explanation of what happens here is presented under 'What really happened?', in the 'So what's going on?' section.

LINKS TO THE EVERYDAY – raindrops, skydivers, high divers, car crashes, cricket bat and ball, billiard balls, playing tennis or golf.

★ Curiosity trap

Take a golf ball and a ping-pong ball (same size, different masses) and drop them from as high as possible into a tray of play dough. You may need to stand on a chair or on a flight of steps. Have students predict which ball will make the deepest indentation in the play dough. Try this again with a marble and a

lead sinker (same size, different masses). Ask students to explain the results.

Gravity pulls more on dense objects, like the golf ball and the sinker, so the force is greater when dense objects hit the play dough and they therefore make a bigger hole.

★ Answers

- a** The crumpling of the car spreads out the time of the collision as well as absorbing some of the energy **b** Seat belts are designed to stretch a little and so reduce the energy your body must absorb **c** Because air is spongy and can be squashed, the bag absorbs lots of energy which would otherwise be transferred to the head of the person
- a** Air resistance and gravity **b** If these two forces are equal and opposite then the raindrop will fall steadily **c** The raindrops in a thunderstorm fall faster. They have a larger air resistance but a much larger weight (pull of gravity)
- When the egg hits the sheet, the sheet moves backwards a little, so the egg is brought to rest over a distance of several centimetres. The force on the eggshell is therefore not large enough to break it. The kinetic energy of the egg is dissipated throughout the sheet (energy is conserved), and the duration of the collision is extended by the elasticity of the sheet (momentum is conserved)
- masses, force, gravity, newtons
- The force of friction always acts in the opposite direction to movement so it tends to slow things down. Opening a parachute increases air resistance (friction), so the object falls more slowly as the air resistance is greater than the pull of gravity. Once the forces balance, the parachutist falls at a lower speed and can control his collision with the Earth so as not to become injured
- a** Air **b** Berthing a boat is a controlled collision with the pontoon. The fenders are made of squishy material

- that absorbs the energy and reduces the force on the boat as its momentum is reversed. **c** The padding absorbs energy like a cushion and therefore reduces the force of impact on your head. **d** If you landed on straight legs the force reflected into your knees would damage your legs; bending them reduces the force they are exposed to
- The gravitational potential energy is changed to kinetic energy. When it hits the wheel with its gear and lever system, it becomes mechanical energy to drive a generator and is then transformed into electrical energy. Energy is not used up, it is just transformed from one form to another
 - a** Newton **b** Kilogram **c** 1 newton **d** You would weigh less as the moon has less mass **e** No **f** 1.5 newtons
 - a** The larger the mass, the greater the distance the car moves **b** The 4WD has more mass, so at similar speeds the 4WD has more momentum than the small car and will do more damage to any object that it strikes **c** The momentum of the 4WD will take longer to change at impact, so the duration of the collision is longer and the force on the occupants will be less
 - a** The planets hurtle through space at speeds that just balance the Sun's gravitational pull, so the Sun's gravity doesn't make them fall into the Sun **b** They are not moving fast enough to escape the Sun's gravity **c** Satellites are moving so fast around the world that gravity never brings them any closer

★ Give us a clue – gathering momentum

What you need

3 sheets of paper the same size and weight, a stopwatch, 3 balls – a basketball, a tennis ball and a ping-pong ball

What to do

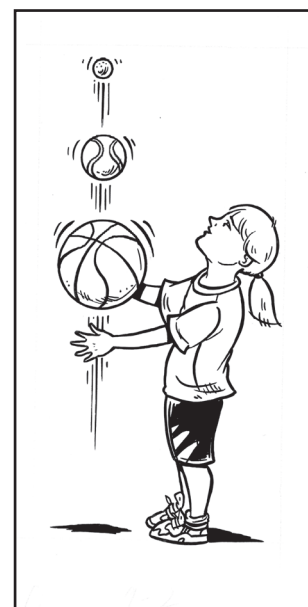
Experiment 1

- 1 **Take** 3 sheets of paper of the same size and weight. **Scrunch** one of them tightly into a small ball. **Fold** the second one until it's a quarter of its original size. Keep the third piece flat.
- 2 Stand in a high position such as a flight of steps or a first floor landing. **Drop** each piece of paper separately. Using the stopwatch, **time** how long it takes for each one to reach the ground.



Experiment 2

- 1 **Drop** the basketball from shoulder height and note how high it bounces.
- 2 **Drop** the tennis ball from the same height and note how high it bounces.
- 3 **Drop** the ping-pong ball from the same height and note how high it bounces.
- 4 Now **hold** the tennis ball on top of the basketball and let both balls go together. Note how high each of the balls bounces this time.
- 5 Now **stack** the ping-pong ball on top of the tennis ball and the tennis ball on top of the basketball. Drop the three-ball stack from the same height.



Now answer these questions:

- Which piece of paper is the densest? Why?
- What force is obstructing the force of gravity and causing the flat piece of paper to take the longest to fall?
- Which ball bounces the highest when each ball is dropped separately?
- Which ball bounces the highest when all three are dropped together? Why?

What happens next?

After you have completed Experiments 1 and 2 and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

Gravity pulls more on dense objects.

Gravitational energy is stored in an object when it is lifted above the surface of the Earth. The further it is lifted, the more energy the object has.

Air resistance is the frictional force or 'drag' on something moving through the air.

Energy cannot be used up or destroyed but it can be transferred from one object to another.

★ The challenge

Design a structure that will allow a raw egg to be dropped from the second floor of a building onto concrete without breaking.

Materials

1 mesh onion bag, 5 pieces of sticky tape (5 cm long), 0.5 m of string, 1 raw egg, 4 large sheets of newspaper, scissors, 1 plastic bag (25 cm × 25 cm)

Instructions

- 1 You may use any of the materials provided that will allow you to successfully construct this structure.
- 2 The onion bag must be on the outside of the structure. If the onion bag rips or if the egg cracks or breaks before the test begins, your team will be disqualified.
- 3 Your design will be tested from first floor level. If the egg remains intact then it will be tested again from the second floor of a building.

Clues!

What is energy? No one really knows, but we do know that nothing in the universe happens without energy. Energy makes things happen. Most of the energy for this planet comes from the Sun.

What is gravity? Gravity is the force of attraction between all objects in the universe. Wherever there is mass there is gravity. The force of gravity is very weak unless one (or both) of the objects has a large mass (e.g. the Earth). The force of gravity on an object is called its **weight**. Force is measured in newtons (N). On Earth, an object with a mass of 100 g will be pulled by gravity with a constant force of 1 newton.

Elastic energy is stored in matter that is able to be stretched, bent or compressed.

Mass is the quantity of matter in an object. Mass is measured in kilograms (kg).

Momentum is the name given to the force with which an object moves.

Newton's Third Law says that for every action there is an equal and opposite reaction.

The **properties** of materials, or how they behave, determine how they are used.

★ So what's going on?

Law of Conservation of Energy: Energy can never be used up or destroyed, but it can be transferred from one object to another or converted from one form to another.

Energy creates all forces, and there are forces involved in every movement in the universe. A **force** is a push or a pull in a particular direction that can change the motion of an object by changing its speed, the direction of its movement or its shape. Heat, sound, elasticity and electricity are all examples of different forms of energy. Gravity, inertia, magnetism, friction and compression are all examples of forces.

Newton's Third Law: For every action there is an equal and opposite reaction.

Forces always occur in pairs, and the two forces are always equal in size but opposite in direction. If these forces are balanced, nothing happens. Only unbalanced forces cause movement. Nothing happens when energy is just stored, either. When an object is pulled down by the force of gravity, it will keep accelerating downwards until it reaches its **terminal velocity** (when the pull of gravity is equal to the push of air resistance). The egg accelerates, or falls faster and faster, towards the Earth. The Earth is far too massive to accelerate towards the egg, so the egg falls and crashes. When the egg lands, the law of action and reaction means that the egg and the concrete push each other with the same force but in opposite directions.

Principle of Conservation of Momentum: When two objects collide, their total momentum remains the same after the collision as it was before.

In other words, when two objects collide, they apply equal and opposite forces to each other. One object may gain momentum and one may lose it but the sum of their individual momentums does not change as long as no other force is acting. **Momentum** is the name given to the force with which an object moves. How fast the momentum (or force) of a moving object changes depends on the size of the force acting on the moving object.

What really happened?

- Using **chemical energy** from the food you have eaten, the structure is lifted above the surface of the Earth to the level of the second floor. It now has stored **gravitational energy** because of its position above the planet. When it is dropped, the gravitational energy becomes the energy of a moving object – called **kinetic energy**. A very small amount of energy is lost as friction through air resistance. When the structure hits the concrete, the concrete exerts an equal and opposite upward force and the structure stops falling. Its **momentum** is reversed.
- The part of the structure that touches the concrete first becomes immobile and the rest of the structure, including the egg, is crushed toward the concrete. As a result, the egg is brought to rest in a very short space of time. The longer the **collision** or change in momentum takes, the smaller the force exerted on the egg inside the device. Some energy is converted to sound energy and the rest is transferred throughout the packaging of the structure and onto the egg.
- The **packaging** around the egg extends the duration of the collision, thus reducing the force on the egg. The packaging should be designed to absorb the kinetic energy and thus reduce the force on the egg to such an extent that its shell is strong enough to resist the final force that reaches it during the collision. Suitable packaging will have **properties** that allow it to be squeezed or forced together, absorbing the kinetic energy as **compression energy**.
- The egg will not break.

★ Figure it out!

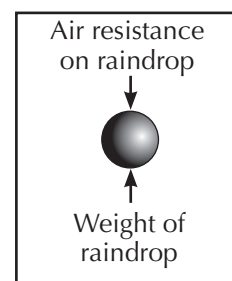
- 1 In any collision, a force is exerted for a length of time. The longer the collision lasts, the smaller the force on the objects that crashed. Explain how the following design features in cars help prevent injury to people involved in a vehicle crash.
 - a The front and back of cars are designed to crumple
 - b seat belts
 - c air bags

- 2 Drops of rain fall down through the air. Study the diagram and answer the questions.

a What two forces act on a falling raindrop?

b If these two forces on the raindrop are balanced, will the raindrop fall at a steady speed?

c Which falls faster – the large drops of a thunderstorm or the small drops when it drizzles? Explain your answer.

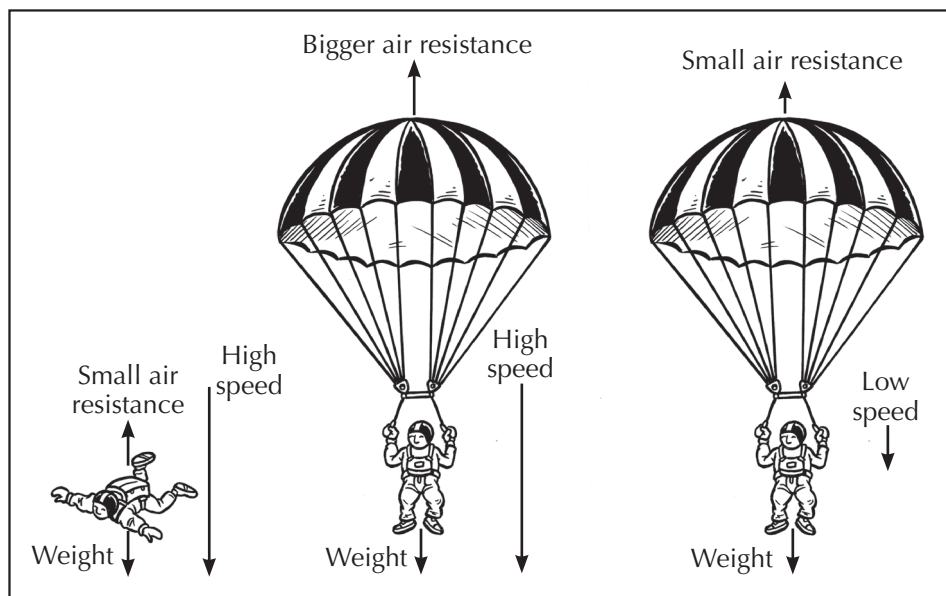


- 3 A group of students carried out the following experiment. Two students held a sheet so that it was stretched tightly across the top and the rest of the sheet hung in a shape like a letter 'J'. Another student threw a raw egg into the vertical part of the sheet and it did not break. Explain what is happening here using the Laws of Conservation of Energy and Momentum.

- 4 Sir Isaac Newton's experiments uncovered one of the fundamental forces in the universe – gravitation. Read the information below and fill in the missing words.

All masses attract all other _____ to a certain extent. The Earth, being a very large mass, exerts a very large _____ of attraction called _____. This force is measured in _____.

- 5 It is not very smart to jump out of an aeroplane – unless you use a parachute. Study the diagram and then explain this statement in terms of balanced forces.



- 6 Answer the following questions in terms of forces and collisions.

a What squishy material do manufacturers put into potato chip bags before they seal them so that the chips do not get broken?

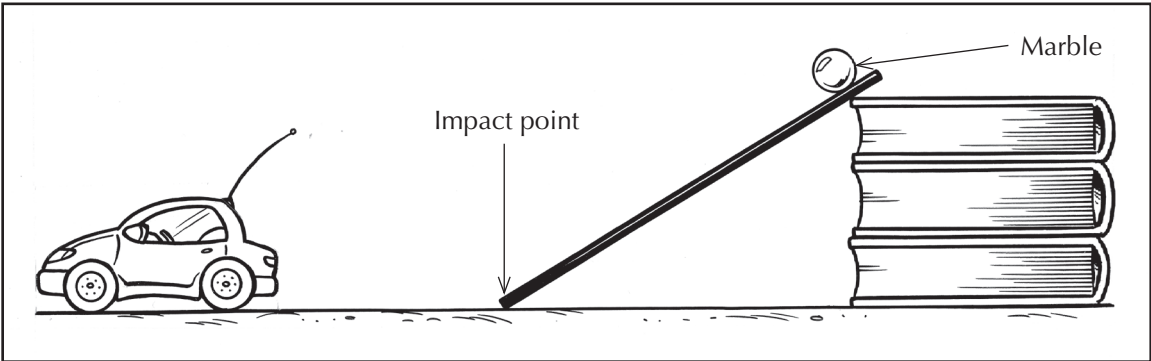
b The captains of boats hang fenders over the sides of the vessel whenever they approach another boat or a berth. Why is this common practice, and what type of material must fenders be made of?

c Safety helmets are worn by bicycle riders, horse riders and racing car drivers. Explain why these helmets are carefully padded inside.

d When you jump from a height onto the ground, why do you bend your knees when you land?

- 7 Water power is based on the fact that water at a higher level has more gravitational potential energy (stored energy) than water at a lower level. If moving water can be made to turn a bladed wheel then electricity can be generated. How does this work in terms of the Law of Conservation of Energy?

- 8 Weight is a very common force. Weight is the force of gravity due to the pull of Earth. The Moon has a smaller mass than the Earth.
- a What unit of measurement is used to measure weight?
- b What unit of measurement is used to measure mass?
- c How much gravitational force pulls on an object that weighs 100 g on Earth?
- d Would you weigh more or less on the moon, compared to on Earth?
- e Would your mass change on the Moon?
- f How much would a 1 kg mass weigh on the moon if the pull of gravity there was only one-sixth of the pull of gravity here?
- 9 An object’s momentum depends on both its mass and its speed. To investigate this idea, an experiment was set up in which a toy car was hit by marbles of different mass. The distance the car moved was recorded in the table below.



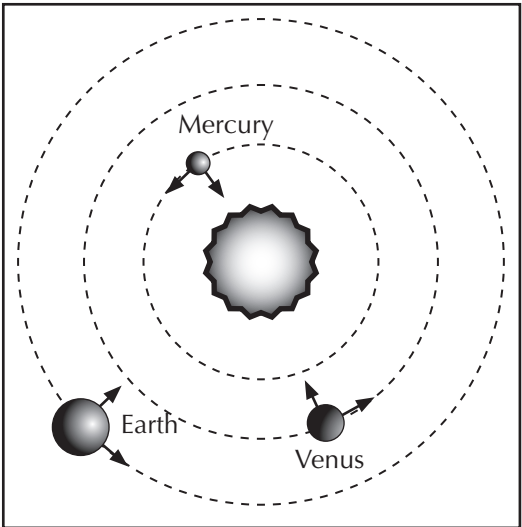
MASS OF THE MARBLE (G)	DISTANCE BETWEEN THE TRACK AND THE REAR BUMPER OF THE CAR (CM)
1	3.5
2	7.6
3	11.5
4	15.8

- a Write a statement relating the distance the car moves to the mass of the marble hitting it.
- b Do these results suggest why, when a large four-wheel drive vehicle travelling at 30 kph hits a small car also travelling at 30 kph, there is more damage to the small car?
- c Why would those travelling in the 4WD be less likely to be seriously injured than those in the small car?

10 Extend your understanding

The Sun has an enormous mass and therefore an enormous amount of gravity. The planets all orbit the Sun. Study the diagram and answer the questions.

- a Why doesn’t the Sun’s gravity make all the planets fall into the Sun?
- b The planets are all moving, so why don’t they move off into space?
- c Man-made artificial satellites orbit the Earth. Why don’t they orbit the Sun like the Earth does?



More information on density can be found in an earlier unit of this book, 'Robust rafts'.

★ Searching for the differences

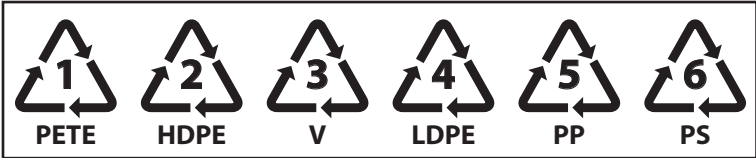
- Rubbing alcohol can be purchased from pharmacies. Prepare the following liquids before the session:

 - a 5:1 mixture of rubbing alcohol and water (5 mL of alcohol to every 1 mL of water)
 - a 3:1 mixture of rubbing alcohol and water (3 mL of alcohol to every 1 mL of water)
 - distilled water
 - 10% salt water (10 g of NaCl to every 90 g of water)
 Each of the five samples in the plastic mixture must be coded. A colour code using permanent pen could be marked on the edge of each sample as follows:

PLASTIC TYPE	DENSITY	OFFICIAL NUMBER CODE	COLOUR CODE
Polyethylene terephthalate (PETE)	1.39	1	red
High density polyethylene (HDPE)	0.95–0.97	2	blue
Vinyl/polyvinyl chloride (V)	1.38	3	purple
Low density polyethylene (LDPE)	0.92–0.94	4	green
Polypropylene (PP)	0.90–0.91	5	black
Polystyrene (PS)	1.05–1.07	6	no colour

★ The challenge

A simple way to separate plastics is to sort them according to density. If an object is dropped into several liquids of different densities, you can work out its density relative to the liquids. The plastics industry has developed a coding system for plastic containers that identifies the type of plastic so they can be sorted. Provide each group of students with samples of the plastics in the table, saved from home and identified by the recycling codes on the bottom of the containers. (Note: do not use 'expanded' versions of plastics as they will all float.) The mystery plastic will be one of these six, but pick one that does not look exactly like the sample of the same number code. Number 2 or number 5 are good



Plastic codes

choices. If the same piece of plastic is used more than once during the testing, it must be dried thoroughly before reuse. Dipping the samples in detergent and water before testing reduces surface tension. The forceps should also be dried well.

LINKS TO THE EVERYDAY – Sports clothing,, soft drink bottles, foam cups, sauce bottles, blister packs for medicines, space craft, credit cards, garden hoses, cars, raincoats, furniture, money.

★ Curiosity trap

You will need two Smiths potato chip packets of the same size and two empty 1.5 L Coke bottles. You can shrink the chip packet before the session begins, but shrink the Coke bottle in front of the students. Flatten an empty chip packet and put it into an oven preheated to 'low'. In a few minutes, the packet will have shrunk to a fraction of the original size. If you use a small chip packet the result is not as impressive,

but a hole can still be punched in the corner and the miniature packet attached to the zipper on a school bag. In front of the students, take one of the Coke bottles, stand it in a container and slowly fill it with boiling water. It will shrink significantly. Shrinking is one of the properties of plastics that can be used to tell them apart; it is also used to heat-seal everything from board games to tooth brushes.

★ Answers

- 1 Carbon, hydrogen, polymers, oil, natural gas, plastics
 2 The PS and PETE will sink: they are denser than water
 3 **a** PVC **b** Flame colour when burnt
 4 **a** Some floating plastics look like food but if swallowed they get stuck in the creatures' stomachs. Six-pack rings get caught on noses, legs and flippers and kill animals **b** Correct sorting and removing impurities
 5 **a** B **b** D **c** C and D **d** A
- 6 **a** 36% **b** 8%
 7 **a** Acrylamide co-polymer **b** White polythene sheeting
 8 The manufacturer grinds up the plastic containers. The vinyl turns to a powder but the contaminants do not. The pure vinyl can be recovered by sieving
 9 **a** Flotation **b** All samples are the same (1 g) **c** PETE **d** Glycerine **e** PP
 10 (d) is correct

Give us a clue – searching for the differences

What you need

A small piece of wax candle, 5 liquids in small containers (a 5:1 mixture of rubbing alcohol and water, a 3:1 mixture of rubbing alcohol and water, distilled water, water with 10% salt, syrup), a small piece of plastic spoon, forceps, paper towels, a mixture of 5 plastics (small samples of codes 1, 2, 4, 5 and 6)

What to do

Experiment 1

- 1 **Arrange** the following 4 liquids in order of density from lowest to highest: salt water, 5:1 rubbing alcohol, distilled water, syrup.
- 2 **Drop** the piece of candle into each of the 4 liquids, drying it well between liquids. In which liquids does it float and in which does it sink? It should sink only in the rubbing alcohol.
- 3 **Now drop** the piece of plastic spoon into each liquid. In which liquids does it sink and in which does it float? It should sink in both the water and the rubbing alcohol.

Experiment 2

The aim is to **separate** the mixture of 5 plastics by taking advantage of the physical property of density. **Record** your results in the data table below.

- 1 **Put** the mixture of 5 plastics into the salt water. Push them down gently with the forceps if nothing sinks immediately. **Skim** off the samples that do not sink and dry them very well.
- 2 **Add** these to the distilled water, skim off the samples that do not sink and dry them.
- 3 **Add** these to the 3:1 rubbing alcohol solution, skim off the ones that do not sink and dry them. Repeat for the 5:1 rubbing alcohol solution and note which plastic does not sink at all.

Data table

LIQUID	DENSITY G/ML	PLASTICS THAT FLOAT
Alcohol and water (5:1)	0.92	
Alcohol and water (3:1)	0.95	
Distilled water	1.00	
Salt water (10%)	1.05	

Now answer the following questions:

- What can you say about the densities of the candle and the spoon?
- Which plastic was the most dense, and which was the least dense?

What happens next?

After you have completed Experiments 1 and 2 and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from these experiments and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

*The **properties** of a substance are what it is like and how it behaves – does it dissolve; how does it burn; does it float?*

***Density** is the mass or number of atoms or molecules in a certain volume of a substance.*

***Mass** is a measure of how much matter, or how many atoms, there are in an object or substance.*

★ The challenge

Within 20 minutes, identify and name the mystery plastic.

Materials

5 small pieces of 6 different plastics, small containers (beakers, or jars) of different liquids (cooking oil, water, rubbing alcohol, a saturated solution of salt water, glycerine), a pencil, 5 samples of a mystery plastic, forceps, paper towels

★ Instructions

- 1 Use the 5 different liquids to test each of the 6 different plastics for density.
- 2 **Record** your results in the data table below by putting a tick in the box if the plastic floats and a cross if it sinks.
- 3 **Test** the mystery plastic.
- 4 **Analyse** all your results and write the full name of the mystery plastic in the space under the data table.

Data table

TYPE OF PLASTIC	VEGETABLE OIL	RUBBING ALCOHOL (3:1)	DISTILLED WATER	GLYCERINE	SALT WATER (10%)
1. PETE					
2. HDPE					
3. V					
4. LDPE					
5. PP					
6. PS					
Mystery					

The mystery plastic is most likely to be _____.

Clues!

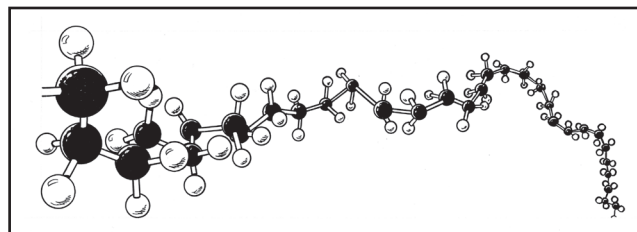
If the density of a substance is less than that of the liquid, it will float. If the density is greater than the liquid, it will sink. Other physical tests for plastics are hardness, bendability and texture.

☆ So what's going on?

Plastic – what is it?

Compounds made only of the atoms carbon and hydrogen are called **hydrocarbons**.

Many plastics are hydrocarbons. The largest and most complicated hydrocarbons are called **polymers** and are mostly man-made from oil and natural gas. Polymers are long chains of identical molecules all joined together like beads on a string. If these chains get tangled up like spaghetti then the plastic is very hard, like polypropylene furniture, but if the chains just keep slipping over one another then they make bendy plastic like polythene soft drink bottles. Plastic fibres like nylon are made by forcing the chains through small holes. Plastics are very common in our world today.



Plastic behaviour

There are a lot of different plastics around but they do not all behave the same way. Chemists have designed plastics to behave in different ways so that we can use them for different jobs.

One way we can separate a mixture is by taking advantage of the physical properties of the individual components of the mixture. Examples of the physical properties of matter include colour, mass, hardness, expansion, shape, the melting points of solids and the boiling points of liquids.

Because some plastics float in water and some do not, flotation can be used to separate mixtures of plastics. Some plastics are more dense than water and some are not.

Density

An important property of a material is how dense it is. The density of water is 1 g/mL/cubic cm. If a plastic floats in water, its density is less than 1 g/cubic cm. If it sinks, its density is more than 1 g/cubic cm. Density can be calculated by dividing the mass (weight) of the substance by its volume. Mass can be measured using a balance, and volume can be measured by displacement or by multiplying the length by the width by the breadth of the object.

Around and around!

The idea of recycling plastics is becoming more and more of an everyday issue. There is growing concern about the environmental impact of plastic garbage. Modern society is generating increasing amounts of landfill, and plastic garbage in particular is endangering marine life. In some communities plastics are separated from other garbage and are recycled. Recycling plastics is more difficult than recycling aluminium cans and paper, because there are many different types of plastics that must be separated before they can be reused. Used plastics can also have foreign residues attached to them, such as labels and food, and these must also be removed.

The manufacture of plastics relies on non-renewable energy sources. Used plastics therefore have a very high energy content, much higher than wood, and could be burned to generate electricity.

The plastics industry has developed a uniform coding system to identify containers by material type and aid in sorting them for recycling purposes. The number code is shown inside a triangular arrow with specific letters underneath. The number and letters indicate the resin (raw chemical) from which the container was made.

Each type of plastic differs in density.



★ Figure it out!

- 1 Read the following information and fill in the missing words.

Compounds made only of the atoms _____ and _____ are called hydrocarbons. Plastics are hydrocarbons. The largest and most complicated hydrocarbons are called _____ and are mostly man-made from _____ and _____. Materials made from these long chains of identical molecules, all joined together, are commonly called _____.

- 2 Five of the most common types of plastics and their densities (g/mL) are listed below:

PP	LDPE	HDPE	PS	PETE/PET
0.90–0.91	0.92–0.94	0.95–0.97	1.05–1.07	1.39

Samples of these five plastics are dropped into water. Predict which ones will sink.

- 3 A group of students separated a mixture of three plastics based on five different properties. Their results are recorded in the following table:

TEST	PVC	PS	PETE/PET
Does it cut or break easily?	With scissors	Breaks	With scissors
Does it bend easily?	Mostly it does	Does not bend	Some do, some don't
Does it float?	No	Yes	Yes
Is it softened by heat?	Yes	Yes	Yes
How does it burn?	Produces white, smelly fumes and a yellow flame	Sooty smoke and a yellow flame, burns easily	Blue flame and not much smoke, burns easily

a An unknown plastic has the following properties: burns with a yellow flame, bends, heat will soften it and it does not float in water. Which of the above plastics is it most likely to be?

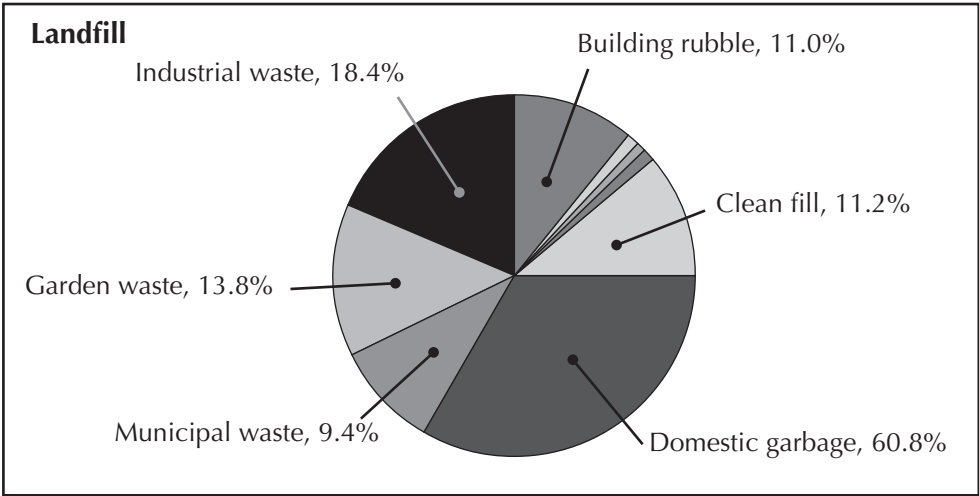
b Which property separates PETE from PVC and PS?

- 4 **a** List two ways that plastic waste is endangering marine life.
b What are two difficulties associated with recycling plastics from domestic sources?
- 5 Study the following table:

OBJECT	DENSITY (kg/m ³)	MASS (kg)	VOLUME (m ³)
A		2000	4
B	8000		4
C		4000	2
D	2000	1000	

- a** Which object has the greatest mass?
- b** Which has the smallest volume?
- c** Which objects could be made of the same substance?
- d** Which object would float on water?

6 Study the graph and answer the questions.



- a What percentage of landfill is domestic garbage?
- b What percentage of domestic garbage is plastic waste?

7 The following table lists the properties of some of the materials that could be used to make disposable nappies.

MATERIAL	PRICE	BIODEGRADABLE	WATERPROOF	WATER ABSORBENCY
Acrylamide co-polymer	medium	yes (slowly)	no	very high
Paper tissue	low	yes	no	medium
Clear polythene sheeting	low	no	yes	nil
White polythene sheeting	low	no	yes	nil

- a Which material would be best used for absorbing nappy waste?
- b Which material would be best used as an outside coating to stop leakage?

8 Manufacturers who use recycled plastics must know that the material is absolutely pure and not contaminated by other plastics. A manufacturer wishes to use recycled vinyl to make new garden hoses. The recycler washes, dries and cools the plastic containers to -114°C . At this temperature the recycled vinyl becomes very brittle but the contaminants stay flexible. How could the manufacturer separate the pure vinyl from the contaminants?

9 Study the following data:

PLASTIC (1 G)	FLOATS IN VEGETABLE OIL	FLOATS IN ALCOHOL AND WATER	FLOATS IN WATER	FLOATS IN GLYCERINE
PETE	No	No	No	No
HDPE	No	No	Yes	Yes
V	No	No	No	No
LDPE	No	Yes	Yes	Yes
PP	Yes	Yes	Yes	Yes
PS	No	No	No	Yes

- a Which physical property is being used to separate these plastics?
- b Which plastic has the most mass?
- c Which plastic has the lowest density?
- d Which liquid is the most dense?
- e Which plastic is denser than all of the liquids?

10 Extend your understanding

BURNS AND FLAMES			
Perpex (PMMA) • Strong smell when burnt	Polystyrene (PS) • Orange flame • Black smoke • Burns easily	Polyethylene (PETE) • Melts and drips • Smells of paraffin • Burns easily	Polyvinyl Chloride (PVC) • Yellow flame • White smoke • Sharp smell

PLASTIC	SMOKE	SMELL
A	Not much	Paraffin
B	White	Strong & stinging
C	None	Strong

- A, B and C, in this order, could be:
- a PVC, PS, PMMA
 - b PETE, PVC, PS
 - c PMMA, PVC, PETE
 - d PETE, PVC, PMMA

Instant Lessons

Quality ready-to-use resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities which may be photocopied for student use. They are unique in providing the highest quality learning outcomes with instant usability.

Versatile uses		Teaching benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
Homework activities	✓	Developing greater student interest and understanding	✓
Reinforcement of learning	✓	A fresh approach to a topic or lesson	✓
Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Problem Solving in Science Book 1

This series helps students to become critical thinkers and independent, creative problem-solvers by stimulating student thinking beyond factual recall and procedural levels. The activities engage students in higher order thinking, which involves the application, synthesis and evaluation of knowledge and skills. For example, students will not only be able to recognise Newton's Laws, but will understand them and be able to apply them to solve new problems.

There is a five-step order of presentation within each of the ten units. Each one begins with a 'Curiosity trap' to grab students' attention and arouse their curiosity. This is followed by the 'Give us a clue' section, a hands-on activity to flesh out the basic scientific notion. Next, 'The challenge' provides the vehicle to apply this notion to an unknown situation. Finally, 'So what's going on?' discusses the theories involved and 'Figure it out!' presents questions to consolidate understanding. Teacher's notes and answers are provided for each unit.

Topics include

Film canister cannons – Newton's 1st and 2nd laws of motion, projectiles and catapults
Reaching for the sky – Bernoulli's Principle and things that fly
Burglar alert – Unlocking the 'mystery' of the electrical circuit
Keep the heat – The kinetic theory of matter in terms of what's 'hot' and what's 'cold'
Robust rafts – Balanced forces, unbalanced forces and Archimedes' Principle

Is it science or is it magic? – The 'lazy' nature of matter and the rules of moving
Under pressure – The atmosphere according to Charles and Boyle
Spaghetti science – Newton, structural forces and the manipulation of materials
Daredevil eggs – Collisions and the conservation of energy and momentum
Plastics to the test – Separating mixtures by searching for the differences

Associated titles in the Instant Lessons series

- Problem Solving in Science (Book 2)
- Scientific Enquiry Skills (Books 1, 2)
- Experimenting with Science
 - The Physical World
 - Forces and Electricity
 - The Living World
- Achieve! Science
 - Chemical Reactions, Materials and Particles
 - Energy, Electricity and Movement
 - Life Processes and the Environment
 - Scientific Investigation, Plants, Rocks and Outer Space

Instant Lessons

An imprint of Blake Education Pty Ltd
ABN 074 266 023
108 Main Rd
Clayton South
VIC 3169
+61 3 9558 4433
www.blake.com.au

ISBN 978-1-92561-006-2



9 781925 610062